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1 Basic

1.1 Dinic [ce2921]

Max flow. $O(V^2E)$; unit caps $O(E\sqrt{E})$; bipartite $O(E\sqrt{V})$. cap is int. 0-base.

```
struct MaxFlow { // 0-base
    struct edge { int to, cap, flow, rev; };
    vector<edge> G[MAXN];
    int s, t, dis[MAXN], cur[MAXN], n;
    int dfs(int u, int cap) {
        if (u == t || !cap) return cap;
        for (int &i = cur[u]; i < (int)G[u].size(); ++i) {
            edge &e = G[u][i];
            if (dis[e.to] == dis[u] + 1 && e.flow != e.cap) {
                int df = dfs(e.to, min(e.cap - e.flow, cap));
                if (df) {
                    e.flow += df;
                    G[e.to][e.rev].flow -= df;
                    return df;
                }
            }
        }
        dis[u] = -1;
        return 0;
    }
    bool bfs() {
        fill_n(dis, n, -1);
        queue<int> q; q.push(s); dis[s] = 0;
        while (!q.empty()) {
            int tmp = q.front(); q.pop();
            for (auto &u : G[tmp])
                if (!dis[u.to] && u.flow != u.cap) {
                    q.push(u.to);
                    dis[u.to] = dis[tmp] + 1;
                }
        }
        return dis[t] != -1;
    }
    int maxflow(int _s, int _t) {
        s = _s, t = _t;
        int flow = 0, df;
        while (bfs()) {
            fill_n(cur, n, 0);
            while ((df = dfs(s, INF))) flow += df;
        }
        return flow;
    }
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i) G[i].clear();
    }
    void reset() {
        for (int i = 0; i < n; ++i)
            for (auto &j : G[i]) j.flow = 0;
    }
    void add_edge(int u, int v, int cap) {
        G[u].pb({v, cap, 0, (int)G[v].size()});
        G[v].pb({u, 0, 0, (int)G[u].size() - 1});
    }
};
```

Usage:
MaxFlow mf; mf.init(n); // nodes 0..n-1
mf.add_edge(u, v, c); // directed
int f = mf.maxflow(s, t);
// dis[i] != -1 => i on s-side of min cut

1.2 vimrc

Editor config. The :Hash commands md5 a visual selection so you can check a typed template against the printed hash.

```
setxkbmap -option ctrl:nocaps
"This file should be placed at ~/.vimrc"
se nu ai hls et ru ic is sc cul re=
1 ts=4 sts=4 sw=4 ls=2 mouse=a
syntax on|hi cursorline cterm=
none ctermbg=89|set bg=dark
inoremap {<CR> {<CR><Esc>ko<tab>
"Select
    region and then type :Hash to hash your selection
    ; useful for verifying that there aren't mistypes."
ca Hash w !cpp -dD -P \|| sed "/^\#.*$/d" \||
tr -d '[:space:]' \|| md5sum \|| cut -c-6
```

```
ca Hash w !cpp -dD -P \|| sed "/^\#.*$/d" \||
tr -d '[:space:]' \|| md5sum \|| cut -c-6
ca Hash w !g++ -E - \|| sed "/^\#.*$/d" \||
tr -d '[:space:]' \|| md5sum \|| cut -c-6
```

1.3 readchar [5f169e]

Buffered getchar via fread_unlocked, amortised $O(1)$. Returns stale data at EOF, so guard the loop.

```
inline char readchar() {
    static char buf[65536], *p = buf, *end = buf;
    if (p == end) end = buf + fread_unlocked(buf,
        1, sizeof buf, stdin), p = buf;
    return *p++;
}
```

Usage:

```
int x = 0; char c = readchar();
while (c < '0' || c > '9') c = readchar();
while (c >= '0' && c <= '9')
    x = x * 10 + c - '0', c = readchar();
```

1.4 BigIntIO [516dcd]

Read/print __int128 (no cin/cout support). Handless sign.

```
bool read(__int128 &value) {
    int ch = getchar(); bool neg;
    while (true) {
        while (ch != EOF && ch != '-' &&
            ch != '+' && !isdigit(ch)) ch = getchar();
        if (ch == EOF) return false;
        neg = ch == '-';
        if (isdigit(ch) || isdigit(ch = getchar())) break;
    }
    __uint128_t x = 0,
        lim = ((__uint128_t)1 << 127) - !neg;
    bool overflow = false;
    do {
        unsigned d = ch - '0';
        if (x > (lim - d) / 10) overflow = true;
        else x = x * 10 + d;
    } while (isdigit(ch = getchar()));
    if (overflow) return false;
    value = !neg ? (__int128)x : x >> 127 ?
        -(__int128)(x - 1) - 1 : -(__int128)x;
    return true;
}

__int128 read() { __int128 value = 0;
    read(value); return value; }
void print_unsigned(__uint128_t x) {
    if (x > 9) print_unsigned(x / 10);
    putchar(x % 10 + '0');
}
void print(__int128 x) {
    if (x < 0) putchar('-');
    print_unsigned(x < 0 ?
        (__uint128_t)(-x + 1) + 1 : x);
}
bool cmp(__int128 x, __int128 y) { return x > y; }
```

Usage:

```
__int128 a = read(), b = read();
print(a * b); putchar('\n');
```

1.5 Black Magic [e66272]

GNU extensions: mt19937; pb_ds red-black tree with find_by_order/order_of_key $O(\log n)$; mergeable heap join; persistent rope. Snippet, not compilable as-is.

```
#include <bits/stdc++.h>
#include <bits/extc++.h>
#include <ext/rope>
using namespace __gnu_pbds; using namespace __gnu_cxx;
typedef tree<int, null_type, std::less<int>,
    rb_tree_tag,
    tree_order_statistics_node_update> tree_set;
typedef cc_hash_table<int, int> umap;
typedef priority_queue<int> heap;
#ifdef BLACK_MAGIC_NO_MAIN
int main() {
    // random
    mt19937 rng(chrono::steady_clock::
        now().time_since_epoch().count());
    auto get_rand = [&](int l, int r) {
        return uniform_int_distribution<int>(l, r)(rng);
    };
    vector<int> v = {1, 2, 3};
    (void)get_rand(0, 10);
    shuffle(v.begin(), v.end(), rng);
    // rb tree
    tree_set s;
    s.insert(71); s.insert(22);
    assert(*s.find_by_order(0) == 22);
    assert(*s.find_by_order(1) == 71);
```

```

assert(s.order_of_key(22) == 0);
assert(s.order_of_key(71) == 1);
s.erase(22);
assert(*s.find_by_order(0) == 71);
assert(s.order_of_key(71) == 0);
// mergable heap
__gnu_pbds::priority_queue<int> a, b;
a.push(1), b.push(2), a.join(b);
// persistent
rope<char> r[2];
r[1] = r[0];
r[1].insert(0, "abc");
r[1].erase(1, 1);
std::cout << r[1].substr(0, 2) << std::endl;
}
#endif

```

1.6 Pragma Optimization [a37289]

Aggressive optimisation and SIMD targets; last line flushes denormals to zero. SIGILL on judges lacking these instruction sets.

```

#pragma GCC optimize("Ofast
, no-stack-protector, no-math-errno, unroll-loops")
#ifdef __x86_64__ || defined(__i386__)
#pragma GCC target("sse, sse2, sse3, sse4")
#pragma GCC target("popcnt, abm, mmx, avx, arch=skylake")
inline void apply_pragmas() {
    __builtin_ia32_ldmxcsr(__builtin_ia32_stmxcsr() |
        0x8040); }
#else
inline void apply_pragmas() {}
#endif

```

1.7 Bitset [e9cd42]

std::bitset API reference. Note: reset() clears to 0 and set() fills with 1 — the comments in the listing are swapped.

```

#include<bits/stdc++.h>
using namespace std;
#ifdef BITSET_NO_MAIN
int main () {
    bitset<4> bit;
    bit.all(); // all bit is true, ret true;
    bit.any(); // any bit is true, ret true
    bit.none(); // all bit is false, ret true
    bit.count();
    bit.to_string('0', '1'); // with parameter
    bit.reset(); // set all to false
    bit.set(); // set all to true
    std::bitset<8> b3{0}, b4{42};
    std::hash<std::bitset<8>> hash_fn8;
    hash_fn8(b3); hash_fn8(b4);
}
#endif

```

2 Graph

2.1 BCC Vertex* [f003fe]

Vertex biconnected components and articulation points, $O(V+E)$. Block-cut tree has up to $2n-2$ nodes — size arrays accordingly. 0-base.

```

struct BCC { // 0-base
    int n, dft, ecnt, nbcc;
    vector<int> low, dfn, bln, stk, is_ap, cir;
    vector<vector<pair<int, int>>> G;
    vector<vector<int>> bcc, nG;
    void make_bcc(int u) { bcc.emplace_back(1, u);
        for (; stk.back() != u; stk.pop_back())
            bln[stk.back()] = nbcc, bcc[nbcc].pb(stk.back());
        stk.pop_back(), bln[u] = nbcc++;
    }
    void dfs(int u, int pe) {
        int child = 0; low[u] = dfn[u] = ++dft, stk.pb(u);
        for (auto [v, e] : G[u])
            if (!dfn[v]) {
                dfs(v, e), ++child;
                low[u] = min(low[u], low[v]);
                if (dfn[u] <= low[v]) {
                    is_ap[u] = 1, bln[u] = nbcc;
                    make_bcc(v), bcc.back().pb(u);
                }
            } else if (e != pe && dfn[v] < dfn[u])
                low[u] = min(low[u], dfn[v]);
        if (pe == -1 && child < 2) is_ap[u] = 0;
        if (pe == -1 && child == 0) make_bcc(u);
    }
    BCC(int _n): n(_n), dft(), ecnt(), nbcc(),
        low(n), dfn(n), bln(n), is_ap(n), G(n) {}
    void add_edge(int u, int v) { G[u].pb(
        v, ecnt), G[v].pb(u, ecnt++); }
}

```

```

void solve() { for (int i = 0; i < n; ++i) if (!dfn[i]) dfs(i, -1); }
void block_cut_tree() {
    int base = nbcc;
    cir.assign(base, 1);
    for (int i = 0; i < n; ++i) if (is_ap[i]) bln[i] = nbcc++;
    nG.assign(nbcc, {});
    for (int i = 0; i < base; ++i) for (int j : bcc[i])
        if (is_ap[j]) nG[i].pb(bln[j]), nG[bln[j]].pb(i);
    } // up to 2 * n - 2 nodes!! bln[i] for id
};

```

Usage:

```

BCC b(n); b.add_edge(u, v); b.solve();
b.bcc[i]; // vertices of i-th block
b.is_ap[u]; b.bln[u];
b.block_cut_tree(); // nG, up to 2n-2 nodes

```

2.2 Bridge* [e040b0]

Bridges and 2-edge-connected components, $O(V+E)$. Uses edge ids rather than the parent vertex, so **multi-edges are handled correctly**. 0-base.

```

struct ECC { // 0-base
    int n, dft, ecnt, necc;
    vector<int> low, dfn, bln, is_bridge, stk;
    vector<vector<pii>> G;
    void dfs(int u, int f) {
        dfn[u] = low[u] = ++dft, stk.pb(u);
        for (auto [v, e] : G[u])
            if (!dfn[v])
                dfs(v, e), low[u] = min(low[u], low[v]);
            else if (e != f)
                low[u] = min(low[u], dfn[v]);
        if (low[u] == dfn[u]) {
            if (f != -1) is_bridge[f] = 1;
            for (; stk.back() != u; stk.pop_back())
                bln[stk.back()] = necc;
            bln[u] = necc++, stk.pop_back();
        }
    }
    ECC(int _n): n(_n), dft(), ecnt(),
        necc(), low(n), dfn(n), bln(n), G(n) {}
    void add_edge(int u, int v) {
        G[u].pb({v, ecnt}), G[v].pb({u, ecnt++});
    }
    void solve() { is_bridge.resize(ecnt);
        for (int i = 0; i < n; ++i) if (!dfn[i]) dfs(i, -1);
    }
}; // ecc_id(i): bln[i]

```

Usage:

```

ECC e(n); e.add_edge(u, v); e.solve();
e.is_bridge[edge_id];
e.bln[u]; // 2-edge-connected comp id
// edge ids -> multi-edges handled right

```

2.3 SCC* [3e1afd]

Tarjan SCC, $O(V+E)$. bln ids come out in **reverse topological order** — 2SAT relies on this. 0-base.

```

struct SCC { // 0-base
    int n, dft, nscc;
    vector<int> low, dfn, bln, instack, stk;
    vector<vector<int>> G;
    void dfs(int u) {
        low[u] = dfn[u] = ++dft, instack[u] = 1, stk.pb(u);
        for (int v : G[u])
            if (!dfn[v])
                dfs(v), low[u] = min(low[u], low[v]);
            else if (instack[v])
                low[u] = min(low[u], dfn[v]);
        if (low[u] == dfn[u]) {
            for (; stk.back() != u; stk.pop_back())
                bln[stk.back()] = nscc,
                    instack[stk.back()] = 0;
            instack[u] = 0, bln[u] = nscc++, stk.pop_back();
        }
    }
    SCC(int _n): n(_n), dft(), nscc(), low(n),
        dfn(n), bln(n), instack(n), G(n) {}
    void add_edge(int u, int v) { G[u].pb(v); }
    void solve() {
        for (int i = 0; i < n; ++i) if (!dfn[i]) dfs(i);
    }
}; // scc_id(i): bln[i]

```

Usage:

```

SCC s(n); s.add_edge(u, v); s.solve();
s.bln[u]; // component id
// ids are in REVERSE topological order

```

2.4 2SAT* [3eaa42]

$O(V+E)$ via implication graph + SCC. Variable i is node i (true) / $i+n$ (false). Requires SCC.cpp. 0-base.

```
struct SAT { // 0-base
    int n;
    vector<bool> istrue;
    SCC scc;
    SAT(int _n): n(_n), istrue(n+n), scc(n+n) {}
    int rv(int a) { return a >= n ? a - n : a + n; }
    void add_clause(int a, int b) {
        scc.add_edge(rv(a), b), scc.add_edge(rv(b), a); }
    bool solve() {
        scc.solve(); for (int i = 0; i < n; ++i)
            if (scc.blm[i] == scc.blm[i+n]) return false;
        else istrue[i] = scc.blm[i] < scc.blm[i+n],
            istrue[i+n] = !istrue[i];
        return true;
    }
};
```

Usage:
 SAT sat(n); // var i true=i, false=i+n
 sat.add_clause(a, b); // (a or b)
 if (sat.solve()) use sat.istrue[i];

2.5 MinimumMeanCycle* [b9cc0e]

Karp's minimum mean cycle, $O(V^3)$ on an adjacency matrix. Returns the mean as a reduced fraction, $(-1, -1)$ if acyclic.

```
ll road[N][N]; // input here
struct MinimumMeanCycle {
    ll dp[N+1][N]; int n;
    static bool less_frac(pll a, pll b) {
        return (___int128)a.first *
            b.second < (___int128)b.first * a.second;
    }
    pll solve() {
        fill_n(dp[0], n, 0);
        // every vertex may be the start
        for (int k = 1; k <= n; ++k) {
            fill_n(dp[k], n, INF);
            for (int v = 0; v < n; ++v)
                for (int u = 0; u < n; ++u)
                    if (dp[k-1][u] < INF && road[u][v] < INF)
                        dp[k][v] = min(dp[k][v],
                            dp[k-1][u] + road[u][v]);
        }
        bool found = false; pll ans;
        for (int v = 0; v < n; ++v) if (dp[n][v] < INF) {
            bool have = false; pll high;
            for (int k = 0; k < n; ++k) if (dp[k][v] < INF) {
                pll cur = {dp[n][v] - dp[k][v], n - k};
                if (!have || less_frac(high, cur)) high = cur, have = true;
            }
            if (have && (!found || less_frac(high, ans))) ans = high, found = true;
        }
        if (!found) return {-1, -1};
        ll g = gcd(ans.first, ans.second);
        return {ans.first / g, ans.second / g};
    }
    void init(int _n) { n = _n; }
};
```

Usage:
 // fill road[i][j], INF if absent
 static MinimumMeanCycle mmc; mmc.init(n);
 auto [p, q] = mmc.solve(); // p/q reduced
 // (-1,-1) if no cycle

2.6 Virtual Tree* [d80ceb]

Compresses k key vertices into an $O(k)$ auxiliary tree in $O(k \log k)$. Needs external LCA(), dfn, dep.

```
vector<int> vG[N];
int top, st[N];
void insert(int u) {
    if (top == -1) return st[++top] = u, void();
    int p = LCA(st[top], u);
    if (p == st[top]) return st[++top] = u, void();
    while (top >= 1 && dep[st[top-1]] >= dep[p])
        vG[st[top-1]].pb(st[top]), --top;
    if (st[top] != p)
        vG[p].pb(st[top]), --top, st[++top] = p;
    st[++top] = u;
}

void reset(int u) {
    for (int i : vG[u]) reset(i); vG[u].clear();
}

int build(vector<int> &v) {
    if (v.empty()) return -1;
    top = -1;
```

```
sort(ALL(v), [&](int a, int b) {
    return dfn[a] < dfn[b]; });
v.erase(unique(ALL(v)), v.end());
for (int i : v) insert(i);
while (top > 0) vG[st[top-1]].pb(st[top]), --top;
return st[0];
}

void solve(vector<int> &v) {
    int root = build(v);
    if (root == -1) return;
    // do something
    reset(root);
}
```

Usage:
 // needs LCA(), dfn[], dep[] prepared
 solve(keys); // builds vG, then resets
 // do the DP where marked in the file

2.7 Maximum Clique Dyn* [d50aa9]

Max clique by branch and bound with dynamic colouring. Exponential, but fast for $n \leq 100$.

```
struct MaxClique { // fast when N <= 100
    bitset<N> G[N], cs[N];
    int ans, sol[N], q, cur[N], d[N], n;
    void init(int _n) {
        n = _n; for (int i = 0; i < n; ++i) G[i].reset();
    }
    void add_edge(int u, int v) { G[u][v] = G[v][u] = 1; }
    void pre_dfs(vector<int> &r, int l, bitset<N> mask) {
        if (l < 4) {
            for (int i : r) d[i] = (G[i] & mask).count();
            sort(ALL(r), [&](int x, int y) {
                return d[x] > d[y]; });
        } vector<int> c(SZ(r));
        int lft = max(ans - q + 1, 1), rgt = 1, tp = 0;
        cs[1].reset(), cs[2].reset();
        for (int p : r) {
            int k = 1;
            while ((cs[k] & G[p]).any()) ++k;
            if (k > rgt) cs[++rgt].reset();
            cs[k][p] = 1;
            if (k < lft) r[tp++] = p;
        } for (int k = lft; k <= rgt; ++k)
            for (int p = cs[k]._Find_first(); p < N;
                p = cs[k]._Find_next(p))
                r[tp] = p, c[tp] = k, ++tp;
        dfs(r, c, l + 1, mask);
    }
    void dfs(vector<int> &r, vector<int> &c,
        int l, bitset<N> mask) {
        while (!r.empty()) {
            int p = r.back();
            r.pop_back(), mask[p] = 0;
            if (q + c.back() <= ans) return;
            cur[q++] = p;
            vector<int> nr;
            for (int i : r) if (G[p][i]) nr.pb(i);
            if (!nr.empty()) pre_dfs(nr, l, mask & G[p]);
            else if (q > ans) ans = q, copy_n(cur, q, sol);
            c.pop_back(), --q;
        }
    }
    int solve() {
        vector<int> r(n); ans = q = 0, iota(ALL(r), 0);
        pre_dfs(r, 0, bitset<N>(string(n, '1')));
        return ans;
    }
};
```

Usage:
 static MaxClique mc; mc.init(n); mc.add_edge(u, v);
 int k = mc.solve(); // sol[0..k] = clique
 // fast for $n \leq 100$

2.8 Minimum Steiner Tree* [8f55a0]

$O(V3^T + V2^{2T})$ for T terminals. Supports vertex costs via vcst; add_edge is directed. 0-base.

```
struct SteinerTree { // 0-base
    int n, dst[N][N], dp[1 << T][N], tdst[N];
    int vcst[N]; // the cost of vertices
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i)
            fill_n(dst[i], n, INF), dst[i][i] = vcst[i] = 0;
    }
    void chmin(int &x, int val) { x = min(x, val); }
    void add_edge(int ui, int vi, int wi) {
        chmin(dst[ui][vi], wi); }
```

```

void shortest_path() {
    // Convert edge-only distances into path costs that
    // include every
    // entered vertex. The source vertex is charged by
    // the singleton state.
    for (int i = 0; i < n; ++i)
        for (int j = 0; j < n; ++j)
            if (i != j &&
                dst[i][j] < INF) dst[i][j] += vcst[j];
    for (int k = 0; k < n; ++k)
        for (int i = 0; i < n; ++i)
            for (int j = 0; j < n; ++j)
                chmin(dst[i][j], dst[i][k] + dst[k][j]);
}

int solve(const vector<int>& ter) {
    vector<int> terminals = ter;
    sort(terminals.begin(), terminals.end());
    terminals.erase(unique(terminals.begin(),
        terminals.end()), terminals.end());
    if (terminals.empty()) return 0;
    shortest_path();
    int t = SZ(terminals), full = (1 << t) - 1;
    for (int i = 0; i <= full; ++i) fill_n(dp[i],
        n, INF);
    copy_n(vcst, n, dp[0]);
    for (int msk = 1; msk <= full; ++msk) {
        if (!(msk & (msk - 1))) {
            int who = __lg(msk);
            for (int i = 0; i < n; ++i)
                dp[msk][i] = vcst[terminals[who]] +
                    dst[terminals[who]][i];
        }
        for (int i = 0; i < n; ++i)
            for (int sub = (msk - 1) & msk;
                sub; sub = (sub - 1) & msk)
                chmin(dp[msk][i],
                    dp[sub][i] + dp[msk ^ sub][i] - vcst[i]);
        for (int i = 0; i < n; ++i) {
            tdst[i] = INF;
            for (int j = 0; j < n; ++j)
                chmin(tdst[i], dp[msk][j] + dst[j][i]);
        }
        copy_n(tdst, n, dp[msk]);
    }
    return *min_element(dp[full], dp[full] + n);
}
}; // O(V 3^T + V^2 2^T)

Usage:
static SteinerTree st; st.init(n); st.vcst[i] = nodeCost;
st.add_edge(u, v, w); // DIRECTED
int c = st.solve(terminals);

```

2.9 Dominator Tree* [68e814]

Lengauer-Tarjan, $O((V + E) \log V)$. Only vertices reachable from root are meaningful. 1-base.

```

struct dominator_tree { // 1-base
    vector<int> G[N], rG[N];
    int n, pa[N], dfn[N], id[N], Time;
    int semi[N], idom[N], best[N];
    vector<int> tree[N]; // dominator_tree
    void init(int _n) {
        n = _n;
        for (int i = 1; i <= n; ++i)
            G[i].clear(), rG[i].clear();
    }
    void add_edge(int u, int v) {
        G[u].pb(v), rG[v].pb(u);
    }
    void dfs(int u) {
        id[dfn[u] = ++Time] = u;
        for (auto v : G[u])
            if (!dfn[v]) dfs(v), pa[dfn[v]] = dfn[u];
    }
    int find(int y, int x) {
        if (y <= x) return y;
        int tmp = find(pa[y], x);
        if (semi[best[y]] > semi[best[pa[y]]]) best[y] =
            best[pa[y]];
        return pa[y] = tmp;
    }
    void tarjan(int root) {
        Time = 0;
        for (int i = 1; i <= n; ++i)
            dfn[i] = idom[i] = 0, tree[i].clear(),
                best[i] = semi[i] = i;
        dfs(root);
        for (int i = Time; i > 1; --i) {
            int u = id[i];
            for (auto v : rG[u])
                if (v = dfn[v]) {
                    find(v, i);
                    semi[i] = min(semi[i], semi[best[v]]);
                }
            tree[semi[i]].pb(i);
        }
    }
}

```

```

    for (auto v : tree[pa[i]]) {
        find(v, pa[i]);
        idom[v] = semi[best[v]] = pa[i] ?
            pa[i] : best[v];
        tree[pa[i]].clear();
    }
    for (int i = 2; i <= Time; ++i) {
        if (idom[i] != semi[i]) idom[i] = idom[idom[i]];
        tree[id[idom[i]]].pb(id[i]);
    }
}
};

Usage:
static dominator_tree
    dt; dt.init(n); dt.add_edge(u, v); // 1-base
dt.tarjan(root);
dt.tree[u]; // children in dominator tree

```

2.10 Minimum Clique Cover* [b8def4]

$O(n2^n)$ probabilistic cover count via subset-sum transform. Needs fwt(); n limited by the 2^n arrays.

```

struct Clique_Cover { // 0-base, O(n2^n)
    int co[1 << N], n, E[N];
    int dp[1 << N];
    void init(int _n) {
        n = _n; fill_n(dp, 1 << n, 0), fill_n(E,
            n, 0), fill_n(co, 1 << n, 0);
    }
    void add_edge(int u, int v) {
        E[u] |= 1 << v, E[v] |= 1 << u;
    }
    int solve() {
        for (int i = 0; i < n; ++i) co[1 << i] = E[i] | (1 << i);
        co[0] = (1 << n) - 1, dp[0] = (n & 1) * 2 - 1;
        for (int i = 1; i < (1 << n); ++i) {
            int t = i & -i;
            dp[i] = -dp[i ^ t], co[i] = co[i ^ t] & co[t];
        }
        for (int i = 0; i < (1 << n); ++i)
            co[i] = (co[i] & i) == i;
        fwt(co, 1 << n, 1);
        for (int ans = 1; ans < n; ++ans) {
            int sum = 0; // probabilistic
            for (int i = 0; i < (1 << n); ++i)
                sum += (dp[i] * co[i]);
            if (sum) return ans;
        }
        return n;
    }
};

Usage:
static Clique_Cover cc; cc.init(n); cc.add_edge(u, v);
int k = cc.solve(); // O(n 2^n)
// probabilistic; needs fwt()

```

2.11 NumberofMaximalClique* [25f6de]

Bron-Kerbosch with pivot, $O(3^{n/3})$. Returns the full count. 1-base.

```

struct BronKerbosch { // 1-base
    int n, S, g[N][N];
    void init(int _n) {
        n = _n;
        for (int i = 1; i <= n; ++i) fill_n(g[i],
            n + 1, 0);
    }
    void add_edge(int u, int v) { g[u][v] = g[v][u] = 1; }
    void dfs_vec(vector<int> P, vector<int> X) {
        if (P.empty() && X.empty()) { ++S; return; }
        int pivot = -1, best = -1;
        vector<int> both = P;
        both.insert(both.end(), X.begin(), X.end());
        for (int u : both) {
            int score = 0;
            for (int v : P) score += g[u][v];
            if (score > best) best = score, pivot = u;
        }
        vector<int> cand;
        for (int v : P) if (pivot == -1 || !g[pivot][v]) cand.push_back(v);
        for (int v : cand) {
            vector<int> nP, nX;
            for (int u : P) if (g[v][u]) nP.push_back(u);
            for (int u : X) if (g[v][u]) nX.push_back(u);
            dfs_vec(nP, nX);
            P.erase(find(P.begin(), P.end(), v));
            X.push_back(v);
        }
    }
    int solve() {
        vector<int> P(n), X; iota(P.begin(), P.end(), 1);
        S = 0, dfs_vec(P, X);
        return S;
    }
}

```

```

    }
};
Usage:
static
BronKerbosch bk; bk.init(n); bk.add_edge(u, v); // 1-base
int c = bk.solve();
// full number of maximal cliques

```

3 Data Structure

3.1 BIT kth* [0002d9]

k -th smallest from a value-domain BIT in $O(\log N)$ by descending the tree. N must be a power of two.

```

int bit[N + 1]; // N = 2 ^ k
int query_kth(int k) {
    int res = 0;
    for (int i = N; i >= 1; i >>= 1)
        if (bit[res + i] < k)
            k -= bit[res + i];
    return res + 1;
}
Usage:
// bit[] is a value-domain BIT, N = 2^k
int v = query_kth(k); // k-th smallest
// O(log N), walks the tree directly

```

3.2 Interval Container* [7916fc]

Maintains disjoint half-open $[L, R)$ intervals with auto-merge; amortised $O(\log n)$.

```

set<pii>::iterator addInterval(set<pii>& is,
    int L, int R) {
    if (L == R) return is.end();
    auto it = is.lower_bound({L, R}); before = it;
    while (it != is.end() && it->X <= R) {
        R = max(R, it->Y);
        before = it = is.erase(it);
    } if (it != is.begin() && (--it)->Y >= L) {
        L = min(L, it->X);
        R = max(R, it->Y);
        is.erase(it);
    } return is.insert(before, {L, R});
}
void removeInterval(set<pii>& is, int L, int R) {
    if (L == R) return;
    auto it = addInterval(is, L, R);
    auto [l2, r2] = *it; is.erase(it);
    if (l2 != L) is.emplace(l2, L);
    if (R != r2) is.emplace(R, r2);
}
Usage:
set<pii> s; // disjoint [L, R)
addInterval(s, 3, 7); // merges overlaps
removeInterval(s, 4, 5); // punches a hole

```

3.3 Leftist Tree [14c121]

Mergeable max-heap, merge/pop $O(\log n)$, keep subtree size and sum. No lazy tags.

```

struct node {
    ll v, data, sz, sum;
    node *l, *r;
    node(ll k) : v(0), data(k),
        sz(1), l(0), r(0), sum(k) {}
};
ll sz(node *p) { return p ? p->sz : 0; }
ll V(node *p) { return p ? p->v : -1; }
ll sum(node *p) { return p ? p->sum : 0; }
node *merge(node *a, node *b) {
    if (!a || !b) return a ? a : b;
    if (a->data < b->data) swap(a, b);
    a->r = merge(a->r, b);
    if (V(a->r) > V(a->l)) swap(a->r, a->l);
    a->v = V(a->r) + 1, a->sz = sz(a->l) + sz(a->r) + 1;
    a->sum = sum(a->l) + sum(a->r) + a->data;
    return a;
}
void pop(node *&o) {
    delete exchange(o, merge(o->l, o->r));
}
Usage:
node *a = new node(x), *b = new node(y); // MAX-heap
a = merge(a, b); // O(log n)
ll top = a->data; pop(a);
// a->sz, a->sum kept per subtree

```

3.4 Dynamic 2D Segment Tree [d930a3]

Rectangle add and rectangle-sum query on half-open coordinates, $O(\log n \log m)$ per operation. Both dimensions allocate nodes dynamically; queries may push lazy tags. Uses int, so large sums overflow.

```

// ----- 1D -----
struct Seg {
    struct Node {
        int sum = 0, lazy = 0;

```

```

        Node *l = nullptr, *r = nullptr;
    };
    int n;
    Node *root = nullptr;
    Seg(int n = 0) : n(n) {}
    void apply(Node *idx, int l, int r, int v) {
        if (!idx) idx = new Node();
        idx->sum += v * (r - l);
        idx->lazy += v;
    }
    void push(Node *idx, int l, int r) {
        if (!idx || !idx->lazy || r - l == 1) return;
        int m = (l + r) / 2;
        apply(idx->l, l, m, idx->lazy);
        apply(idx->r, m, r, idx->lazy);
        idx->lazy = 0;
    }
    void update(Node
        *idx, int l, int r, int ql, int qr, int v) {
        if (qr <= l || r <= ql) return;
        if (!idx) idx = new Node();
        if (ql <= l && r <= qr) {
            apply(idx, l, r, v);
            return;
        }
        push(idx, l, r);
        int m = (l + r) / 2;
        update(idx->l, l, m, ql, qr, v);
        update(idx->r, m, r, ql, qr, v);
        idx->sum =
            (idx->l ? idx->l->sum : 0) +
            (idx->r ? idx->r->sum : 0);
    }
    int query
        (Node *idx, int l, int r, int ql, int qr) {
        if (!idx || qr <= l || r <= ql) return 0;
        if (ql <= l && r <= qr) return idx->sum;
        push(idx, l, r);
        int m = (l + r) / 2;
        return query(idx->l, l, m, ql, qr)
            + query(idx->r, m, r, ql, qr);
    }
    void update(int l, int r, int v) {
        update(root, 0, n, l, r, v);
    }
    int query(int l, int r) {
        return query(root, 0, n, l, r);
    }
};
// ----- 2D -----
struct Seg2D {
    struct Node {
        Seg sum, lazy;
        Node *l = nullptr, *r = nullptr;
        Node(int m) : sum(m), lazy(m) {}
    };
    int n, m;
    Node *root = nullptr;
    Seg2D(int n, int m) : n(n), m(m) {}
    void update(Node *idx, int l, int r,
        int xl,
        int xr, int yl, int yr, int v) {
        if (xr <= l || r <= xl) return;
        if (!idx) idx = new Node(m);
        int rows = min(r, xr) - max(l, xl);
        idx->sum.update(yl, yr, v * rows);
        if (xl <= l && r <= xr) {
            idx->lazy.update(yl, yr, v);
            return;
        }
        int md = (l + r) / 2;
        update(idx->l, l, md, xl, xr, yl, yr, v);
        update(idx->r, md, r, xl, xr, yl, yr, v);
    }
    int query(Node *idx, int l, int r,
        int xl, int xr, int yl, int yr) {
        if (!idx || xr <= l || r <= xl) return 0;
        if (xl <= l && r <= xr)
            return idx->sum.query(yl, yr);
        int rows = min(r, xr) - max(l, xl);
        int md = (l + r) / 2;
        return idx->lazy.query(yl, yr) * rows
            + query(idx->l, l, md, xl, xr, yl, yr)
            + query(idx->r, md, r, xl, xr, yl, yr);
    }
    void

```



```

        update(int xl, int xr, int yl, int yr, int v) {
            update(root, 0, n, xl, xr, yl, yr, v);
        }
        int query(int xl, int xr, int yl, int yr) {
            return query(root, 0, n, xl, xr, yl, yr);
        }
    };
Usage:
Seg2D st(n, m); // coordinates are half-open
st.update(xl, xr, yl, yr, v); // rectangle add
int sum = st.query(xl, xr, yl, yr); // rectangle sum

```

3.5 Heavy light Decomposition* [2e43dc]

Vertex path sums in $O(\log^2 n)$ using Seg from the preceding template. Tree vertices are 1-base, root 1; flattened positions are half-open and 0-base.

```

struct Heavy_light_Decomposition { // 1-base; needs Seg
    int n, ulink[N], deep[N], mxson[N], w[N], pa[N];
    int t, pl[N], val[N]; // val: vertex data
    Seg seg;
    vector<int> G[N];
    void init(int _n) {
        n = _n;
        w[0] = 0;
        for (int i = 1; i <= n; ++i)
            G[i].clear(), mxson[i] = 0;
    }
    void add_edge(int a, int b) { G[a].pb(b), G[b].pb(a); }
    void dfs(int u, int f, int d) {
        w[u] = 1, pa[u] = f, deep[u] = d++;
        for (int &i : G[u])
            if (i != f)
                dfs(i, u, d), w[u] += w[i],
                    mxson[u] = w[mxson[u]] < w[i] ? i : mxson[u];
    }
    void cut(int u, int link) {
        pl[u] = t++; ulink[u] = link;
        seg.update(pl[u], pl[u] + 1, val[u]);
        if (!mxson[u]) return;
        cut(mxson[u], link);
        for (int i : G[u])
            if (i != pa[u] && i != mxson[u]) cut(i, i);
    }
    void build() { t = 0, seg = Seg(n), dfs(1, 1, 1), cut(1, 1); }
    int query(int a, int b) {
        int ta = ulink[a], tb = ulink[b], res = 0;
        while (ta != tb) {
            if (deep[ta] > deep[tb]) swap(ta, tb), swap(a, b);
            res += seg.query(pl[tb], pl[b] + 1);
            tb = ulink[b = pa[tb]];
        } if (pl[a] > pl[b]) swap(a, b);
        return res + seg.query(pl[a], pl[b] + 1);
    }
};
Usage:
static Heavy_light_Decomposition
hld; hld.init(n); hld.add_edge(a, b);
hld.val[u] = w; hld.build(); // 1-base
hld.query(a, b); // O(log^2 n)
// skeleton: wire a segtree into build/query

```

3.6 Centroid Decomposition* [555d05]

Builds a centroid tree in $O(n \log n)$; modify/query are $O(\log n)$. upinfo removes same-branch overcount. 1-base.

```

struct Cent_Dec { // 1-base
    vector<pll> G[N];
    pll info[N]; // store info. of itself
    pll upinfo[N]; // store info. of climbing up
    int n, pa[N], layer[N], sz[N], done[N];
    ll dis[lg(N) + 1][N];
    void init(int _n) {
        n = _n, layer[0] = -1;
        fill_n(pa + 1, n, 0), fill_n(done + 1, n, 0);
        fill_n(info, n + 1, pll());
        fill_n(upinfo, n + 1, pll());
        for (int i = 1; i <= n; ++i) G[i].clear();
    }
    void add_edge(int a, int b, int w) {
        G[a].pb(pll(b, w)), G[b].pb(pll(a, w));
    }
    void get_cent(int u, int f,
        int &mx, int &c, int num) {
        int mxsz = 0; sz[u] = 1;
        for (pll e : G[u])
            if (!done[e.X] && e.X != f) {
                get_cent(e.X, u, mx, c, num);
                sz[u] += sz[e.X], mxsz = max(mxsz, sz[e.X]);
            } if (mx > (mxsz = max(mxsz, num - sz[u])))

```

```

        mx = mxsz, c = u;
    }
    void dfs(int u, int f, ll d, int org) {
        // if required, add self info or climbing info
        dis[layer[org]][u] = d;
        for (pll e : G[u])
            if (!done[e.X] && e.X != f)
                dfs(e.X, u, d + e.Y, org);
    }
    int cut(int u, int f, int num) {
        int mx = 1e9, c = 0, lc;
        get_cent(u, f, mx, c, num);
        done[c] = 1, pa[c] = f, layer[c] = layer[f] + 1;
        dis[layer[c]][c] = 0;
        for (pll e : G[c])
            if (!done[e.X]) {
                lc = sz[e.X] > sz[c] ? cut(e.X, c,
                    num - sz[c]) : cut(e.X, c, sz[e.X]);
                upinfo[lc] = pll(), dfs(e.X, c, e.Y, c);
            } return done[c] = 0, c;
    }
    void build() {
        cut(1, 0, n);
    }
    void modify(int u) {
        for (int a = u,
            ly = layer[a]; a; a = pa[a], --ly) {
            info[a].X += dis[ly][u], ++info[a].Y;
            if (pa[a])
                upinfo[a].X += dis[ly - 1][u], ++upinfo[a].Y;
        }
    }
    ll query(int u) {
        ll rt = 0;
        for (int a = u,
            ly = layer[a]; a; a = pa[a], --ly) {
            rt += info[a].X + info[a].Y * dis[ly][u];
            if (pa[a])
                rt -= upinfo[a].X +
                    upinfo[a].Y * dis[ly - 1][u];
        } return rt;
    }
};
Usage:
static Cent_Dec cd; cd.init(n); cd.add_edge(a, b, w);
cd.build(); // 1-base
cd.modify(u); // mark u
ll s = cd.query(u); // sum dist to marked

```

3.7 LiChaoST* [8d3b73]

Line container over an integer x domain, $O(\log n)$. Keeps the **maximum**; negate for minimum.

```

struct L {
    ll m, k, id;
    L() : id(-1) {}
    L(ll a, ll b, ll c) : m(a), k(b), id(c) {}
    L at(ll x) { return m * x + k; }
};
class LiChao { // maintain max
    int n; vector<L> nodes;
    void insert(int l, int r, int rt, L ln) {
        int m = (l + r) >> 1;
        L &o = nodes[rt];
        if (o.id == -1) return o = ln, void();
        bool a = o.at(l) < ln.at(l);
        if (o.at(m) < ln.at(m)) a ^= 1, swap(o, ln);
        if (r - l == 1) return;
        if (a) insert(l, m, rt << 1, ln);
        else insert(m, r, rt << 1 | 1, ln);
    }
    ll query(int l, int r, int rt, ll x) {
        int m = (l + r) >> 1; ll ret = -INF;
        if (nodes[rt].id != -1) ret = nodes[rt].at(x);
        if (r - l == 1) return ret;
        if (x < m) return max(ret,
            query(l, m, rt << 1, x));
        return max(ret, query(m, r, rt << 1 | 1, x));
    } public:
    LiChao(int n_) : n(n_), nodes(n * 4) {}
    void insert(L ln) { insert(0, n, 1, ln); }
    ll query(ll x) { return query(0, n, 1, x); }
};
Usage:
LiChao lc(n); // x domain [0, n)
lc.insert(L(m, k, id)); // y = m*x + k
ll best = lc.query(x); // MAXIMUM
// for min: insert negated, negate result

```

3.8 XOR Basis [5e9147]

Inserts a 64-bit unsigned integer into a linear basis in $O(64)$. `basis[i]` has highest set bit `i`; a dependent value is discarded.

```
ull basis[64];
```

```
void ins(ull x) {
    for (int i = 63; i >= 0; i--)
        if (x >> i & 1) {
            if (!basis[i]) return void(basis[i] = x);
            x ^= basis[i];
        }
}
```

Usage:

```
ins(x); // add x to the XOR basis
```

3.9 KDTree [25028f]

Nearest neighbour, build $O(n \log n)$, query $O(\log n)$ average. Return squared distance and skips distance-0 points.

```
namespace kdt {
    int root, lc[1maxn], rc[1maxn], xl[1maxn], xr[1maxn],
        yl[1maxn], yr[1maxn];
    point p[1maxn];
    int build(int l, int r, int dep = 0) {
        if (l == r) return -1;
        function<bool(const point &, const point &> f =
            [dep](const point &a, const point &b) {
                return dep & 1 ? a.x < b.x : a.y < b.y;
            });
        int m = (l + r) >> 1;
        nth_element(p + l, p + m, p + r, f);
        xl[m] = xr[m] = p[m].x;
        yl[m] = yr[m] = p[m].y;
        lc[m] = build(l, m, dep + 1);
        if (~lc[m])
            xl[m] = min(xl[m], xl[lc[m]]),
            xr[m] = max(xr[m], xr[lc[m]]),
            yl[m] = min(yl[m], yl[lc[m]]),
            yr[m] = max(yr[m], yr[lc[m]]);
        rc[m] = build(m + 1, r, dep + 1);
        if (~rc[m])
            xl[m] = min(xl[m], xl[rc[m]]),
            xr[m] = max(xr[m], xr[rc[m]]),
            yl[m] = min(yl[m], yl[rc[m]]),
            yr[m] = max(yr[m], yr[rc[m]]);
        return m;
    }
    bool bound(const point &q, int o, long long d) {
        double ds = sqrt(d + 1.0);
        return !(q.x < xl[o] - ds || q.x > xr[o] + ds ||
            q.y < yl[o] - ds || q.y > yr[o] + ds);
    }
    long long dist(const point &a, const point &b) {
        return (a.x - b.x) * 1ll * (a.x - b.x) +
            (a.y - b.y) * 1ll * (a.y - b.y);
    }
    void dfs(
        const point &q, long long &d, int o, int dep = 0) {
        if (!bound(q, o, d)) return;
        long long cd = dist(p[o], q);
        if (cd) d = min(d, cd);
        int a = lc[o], b = rc[o];
        if (!(dep & 1 ? q.x < p[o].x : q.y < p[o].y)) swap(a, b);
        if (~a) dfs(q, d, a, dep + 1);
        if (~b) dfs(q, d, b, dep + 1);
    }
    void init(const vector<point> &v) {
        copy(v.begin(), v.end(), p);
        root = build(0, v.size());
    }
    long long nearest(const point &q) {
        long long res = 1e18;
        if (~root) dfs(q, res, root); return res;
    }
} // namespace kdt
```

Usage:

```
kdt::init(pts);
long long d2 = kdt::nearest(q);
// squared distance; SKIPS points at
// distance 0 (i.e. excludes q itself)
```

3.10 Treap [e99c54]

Rotation-free treap, expected $O(\log n)$; split by key, split2 by size. `down()` is empty — add lazy tags there.

```
struct node {
    int data, sz; node *l, *r;
    node(int k) : data(k), sz(1), l(0), r(0) {}
    void up() {
        sz = 1 + (l ? l->sz : 0) + (r ? r->sz : 0);
    }
}
```

```

    }
    void down() {}
};
int sz(node *a) { return a ? a->sz : 0; }
node *merge(node *a, node *b) {
    if (!a || !b) return a ? a : b;
    if (rand() % (sz(a) + sz(b)) < sz(a))
        return a->down(),
            a->r = merge(a->r, b), a->up(), a;
    return b->down(), b->l = merge(a, b->l), b->up(), b;
}
void split(node *o, node *&a, node *&b, int k) {
    if (!o) return a = b = 0, void();
    o->down();
    if (o->data <= k)
        a = o, split(o->r, a->r, b, k), a->up();
    else b = o, split(o->l, a, b->l, k), b->up();
}
void split2(node *o, node *&a, node *&b, int k) {
    if (sz(o) <= k) return a = o, b = 0, void();
    o->down();
    if (sz(o->l) < k)
        a = o, split2(o->r, a->r, b, k - sz(o->l) - 1);
    else b = o, split2(o->l, a, b->l, k);
    o->up();
}
node *kth(node *o, int k) {
    int s = sz(o->l);
    if (k <= s) return kth(o->l, k);
    if (k == ++s) return o;
    return kth(o->r, k - s);
}
int Rank(node *o, int key) {
    if (!o) return 0;
    if (o->data < key)
        return sz(o->l) + 1 + Rank(o->r, key);
    return Rank(o->l, key);
}
bool erase(node *&o, int k) {
    if (!o) return 0;
    if (o->data == k) {
        node *t = o;
        o->down(), o = merge(o->l, o->r);
        delete t;
        return 1;
    }
    node *&t = k < o->data ? o->l : o->r;
    return erase(t, k) ? o->up(), 1 : 0;
}
void insert(node *&o, int k) {
    node *a, *b;
    split(o, a, b, k), o = merge(a,
        merge(new node(k), b));
}
void interval(node *&o, int l, int r) {
    node *a, *b, *c;
    split2(o, a, b, l - 1), split2(b, b, c, r);
    o = merge(a, merge(b, c));
}
Usage:
node *root = nullptr; insert(root, k); erase(root, k);
node *t = kth(root, i); // i-th, 1-base
int r = Rank(root, key); // # less than key
interval(root, l, r); // split2 by size
```

4 Flow/Matching

4.1 Bipartite Matching* [784535]

Max bipartite matching, Hopcroft-Karp. $O(E\sqrt{V})$. 0-base.

```
struct Bipartite_Matching { // 0-base
    int mp[N], mq[N], dis[N + 1], cur[N], l, r;
    vector<int> G[N + 1];
    bool dfs(int u) {
        for (int &i = cur[u]; i < SZ[G[u]]; ++i) {
            int e = G[u][i];
            if (mq[e] == 1 ||
                (dis[mq[e]] == dis[u] + 1 && dfs(mq[e])))
                return mp[mq[e] = u] = e, 1;
        }
        return dis[u] = -1, 0;
    }
    bool bfs() {
        queue<int> q; fill_n(dis, l + 1, -1);
        for (int i = 0; i < l; ++i)
            if (!~mp[i]) q.push(i), dis[i] = 0;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int e : G[u])
                if (!~dis[mq[e]])
```

```

    q.push(mq[e]), dis[mq[e]] = dis[u] + 1;
} return dis[l] != -1;
}
int matching() {
    int res = 0; fill_n(mp, l, -1), fill_n(mq, r, 1);
    while (bfs()) {
        fill_n(cur, l, 0);
        for (int i = 0; i < l; ++i)
            res += (!mp[i] && dfs(i));
    } return res;
}
void add_edge(int s, int t) { G[s].pb(t); }
void init(int _l, int _r) {
    l = _l, r = _r;
    for (int i = 0; i <= l; ++i) G[i].clear();
}
};
Usage:
static Bipartite_Matching bm; bm.init(nl, nr); // left 0..nl-1
bm.add_edge(l, r);
int m = bm.matching();
// pair (i, bm.mp[i]) when bm.mp[i] != -1

```

4.2 Kuhn Munkres* [4b3863]

Max weight *perfect* matching on a square bipartite graph. $O(V^3)$. 0-base.

```

struct KM { // 0-base, maximum matching
    ll w[N][N], hl[N], hr[N], slk[N];
    int fl[N], fr[N], pre[N], qu[N], ql, qr, n;
    bool vl[N], vr[N];
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i) fill_n(w[i], n, -INF);
    }
    void add_edge(int a, int b, ll wei) { w[a][b] = wei; }
    bool Check(int x) {
        if (vl[x] == 1, ~fl[x])
            return vr[qu[qr++] = fl[x]] = 1;
        while (~x) swap(x, fr[fl[x] = pre[x]]);
        return 0;
    }
    void bfs(int s) {
        fill_n(slk, n, INF), fill_n(vl, n, 0), fill_n(vr, n, 0);
        ql = qr = 0, qu[qr++] = s, vr[s] = 1;
        for (ll d;;) {
            while (ql < qr)
                for (int x = 0, y = qu[ql++]; x < n; ++x)
                    if (!vl[x] && slk[x] >= (d = hl[x] + hr[y] - w[x][y])) {
                        if (pre[x] == y, d) slk[x] = d;
                        else if (!Check(x)) return;
                    }
                d = INF;
            for (int x = 0; x < n; ++x)
                if (!vl[x] && d > slk[x]) d = slk[x];
            for (int x = 0; x < n; ++x) {
                if (vl[x]) hl[x] += d;
                else slk[x] -= d;
                if (vr[x]) hr[x] -= d;
            }
            for (int x = 0; x < n; ++x)
                if (!vl[x] && !slk[x] && !Check(x)) return;
        }
    }
    ll solve() {
        fill_n(fl, n, -1), fill_n(fr, n, -1), fill_n(hr, n, 0);
        for (int i = 0; i < n; ++i)
            hl[i] = *max_element(w[i], w[i] + n);
        for (int i = 0; i < n; ++i) bfs(i);
        ll res = 0;
        for (int i = 0; i < n; ++i) res += w[i][fl[i]];
        return res;
    }
};
Usage:
static KM km; km.init(n); // square; missing edge = -INF
km.add_edge(i, j, w); // min: use -w
ll best = km.solve();
// km.fl[j] = left node matched to right j

```

4.3 MincostMaxflow* [2a8a18]

Mincost maxflow, SPFA + Johnson potentials (handles negative cost). 0-base.

```

struct MinCostMaxFlow { // 0-base
    struct Edge
    { ll from, to, cap, flow, cost, rev; } *past[N];
    vector<Edge> G[N];
    int inq[N], n, s, t;
    ll dis[N], up[N], pot[N];

```

```

    bool BellmanFord() {
        fill_n(dis, n, INF), fill_n(inq, n, 0);
        queue<int> q;
        auto relax = [&](int u, ll d, ll cap, Edge *e) {
            if (cap > 0 && dis[u] > d) {
                dis[u] = d, up[u] = cap, past[u] = e;
                if (!inq[u]) inq[u] = 1, q.push(u);
            }
        }; relax(s, 0, INF, 0);
        while (!q.empty()) {
            int u = q.front(); q.pop(), inq[u] = 0;
            for (auto &e : G[u])
                relax(e.to,
                    dis[u] + e.cost + pot[u] - pot[e.to],
                    min(up[u], e.cap - e.flow), &e);
        } return dis[t] != INF;
    }
    void solve(int _s, int _t, ll &flow, ll &cost,
        bool neg = true) {
        s = _s, t = _t, flow = 0, cost = 0;
        if (neg) BellmanFord(), copy_n(dis, n, pot);
        for (; BellmanFord(); copy_n(dis, n, pot)) {
            for (int i = 0; i < n; ++i)
                dis[i] += pot[i] - pot[s];
            flow += up[t], cost += up[t] * dis[t];
            for (int i = t; past[i]; i = past[i]->from) {
                auto &e = *past[i];
                e.flow += up[t], G[e.to][e.rev].flow -= up[t];
            }
        }
        void init(int _n) {
            n = _n, fill_n(pot, n, 0);
            for (int i = 0; i < n; ++i) G[i].clear();
        }
        void add_edge(ll a, ll b, ll cap, ll cost) {
            G[a].pb({a, b, cap, 0, cost, SZ(G[b])});
            G[b].pb({b, a, 0, 0, -cost, SZ(G[a]) - 1});
        }
    };
    Usage:
    static MinCostMaxFlow mcmf; mcmf.init(n);
    mcmf.add_edge(u, v, cap, cost);
    ll flow, cost;
    mcmf.solve(s, t, flow, cost); // neg=true handles negative edges

```

4.4 Maximum Simple Graph Matching*

[c16056]

Max matching on a general graph (blossom). $O(V^3)$. 0-base.

```

struct Matching { // 0-base
    queue<int> q; int n;
    vector<int> fa, s, vis, pre, match;
    vector<vector<int>>> G;
    int Find(int u) {
        return u == fa[u] ? u : fa[u] = Find(fa[u]);
    }
    int LCA(int x, int y) {
        static int tk = 0; tk++; x = Find(x); y = Find(y);
        for (; swap(x, y) if (x != n) {
            if (vis[x] == tk) return x;
            vis[x] = tk;
            x = Find(pre[match[x]]);
        }
    }
    void Blossom(int x, int y, int l) {
        for (; Find(x) != l; x = pre[y]) {
            pre[x] = y, y = match[x];
            if (s[y] == 1) q.push(y), s[y] = 0;
            for (int z : {x, y}) if (fa[z] == z) fa[z] = l;
        }
    }
    bool Bfs(int r) {
        iota(ALL(fa), 0); fill(ALL(s), -1);
        q = queue<int>(); q.push(r); s[r] = 0;
        for (; !q.empty(); q.pop()) {
            int x = q.front(); for (int u : G[x])
                if (s[u] == -1) {
                    if (pre[u] == x, s[u] == 1, match[u] == n) {
                        for (int a = u, b = x, last;
                            b != n; a = last, b = pre[a])
                            last = match[b],
                                match[b] = a, match[a] = b;
                        return true;
                    }
                    q.push(match[u]); s[match[u]] = 0;
                }
            else if (!s[u] && Find(u) != Find(x)) {
                int l = LCA(u, x);
                Blossom(x, u, l); Blossom(u, x, l);
            }
        }
    }

```



```

    }
} return false;
}
Matching(int _n) : n(_n), fa(n + 1), s(n + 1),
    vis(n + 1), pre(n + 1, n), match(n + 1, n), G(n) {}
void add_edge(int u, int v) { G[u].pb(v), G[v].pb(u);
}
int solve() {
    int ans = 0;
    for (int x = 0; x < n; ++x)
        if (match[x] == n) ans += Bfs(x);
    return ans;
}
};
Usage:
Matching M(n); // general graph
M.add_edge(u, v);
int sz = M.solve();
// M.match[x] == n => x is unmatched

```

4.5 Maximum Weight Matching* [dd4d97]

Max weight matching on a general graph (weighted blossom). $O(V^3)$. 1-base.

```

#define REP(i, l, r) for (int i=l; i<=(r); ++i)
struct WeightGraph { // 1-based
    struct edge { int u, v, w; }; int n, nx;
    vector<int> lab; vector<vector<edge>> g;
    vector<int> slk, match, st, pa, S, vis;
    vector<vector<int>> flo, flo_from; queue<int> q;
    WeightGraph(int _n) : n(_n), nx(n * 2), lab(nx + 1),
        g(nx + 1, vector<edge>(nx + 1)), slk(nx + 1),
        flo(nx + 1), flo_from(nx + 1, vector(n + 1, 0)) {
        match = st = pa = S = vis = slk;
        REP(u, 1, n) REP(v, 1, n) g[u][v] = {u, v, 0};
    }
    int E(edge e)
    { return lab[e.u] + lab[e.v] - g[e.u][e.v].w * 2; }
    void update_slk(int u, int x, int &s)
    { if (!s || E(g[u][x]) < E(g[s][x])) s = u; }
    void set_slk(int x) {
        slk[x] = 0; REP(u, 1, n)
            if (g[u][x].w > 0 && st[u] != x && S[st[u]] == 0)
                update_slk(u, x, slk[x]);
    }
    void q_push(int x) {
        if (x <= n) q.push(x);
        else for (int y : flo[x]) q_push(y);
    }
    void set_st(int x, int b) {
        st[x] = b;
        if (x > n) for (int y : flo[x]) set_st(y, b);
    }
    vector<int> split_flo(vector<int> &f, int xr) {
        auto it = find(ALL(f), xr);
        if (auto pr = it - f.begin(); pr % 2 == 1)
            reverse(1 + ALL(f), it = f.end() - pr);
        auto res = vector(f.begin(), it);
        return f.erase(f.begin(), it), res;
    }
    void set_match(int u, int v) {
        match[u] = g[u][v].v;
        if (u <= n) return;
        int xr = flo_from[u][g[u][v].u];
        auto &f = flo[u], z = split_flo(f, xr);
        REP(i, 0, SZ(z) - 1) set_match(z[i], z[i ^ 1]);
        set_match(xr, v); f.insert(f.end(), ALL(z));
    }
    void augment(int u, int v) {
        for (;;) {
            int xnv = st[match[u]]; set_match(u, v);
            if (!xnv) return;
            set_match(v = xnv, u = st[pa[xnv]]);
        }
    }
    int lca(int u, int v) {
        static int t = 0;
        for (++t; u || v; swap(u, v)) if (u) {
            if (vis[u] == t) return u;
            vis[u] = t, u = st[match[u]];
            if (u) u = st[pa[u]];
        } return 0;
    }
    void add_blossom(int u, int o, int v) {
        int b = find(n + 1 + ALL(st), 0) - begin(st);
        lab[b] = 0, S[b] = 0, match[b] = match[o];
        vector<int> f = {o};
        for (int t : {u, v}) {
            reverse(1 + ALL(f));

```

```

            for (int x = t, y; x != o; x = st[pa[y]])
                f.pb(x), f.pb(y = st[match[x]]), q_push(y);
        } flo[b] = f; set_st(b, b);
        REP(x, 1, nx) g[b][x].w = g[x][b].w = 0;
        fill(ALL(flo_from[b]), 0);
        for (int xs : flo[b]) {
            REP(x, 1, nx)
                if (g[b][x].w == 0 || E(g[xs][x]) < E(g[b][x]))
                    g[b][x] = g[xs][x], g[x][b] = g[x][xs];
            REP(x, 1, n)
                if (flo_from[xs][x]) flo_from[b][x] = xs;
        } set_slk(b);
    }
    void expand_blossom(int b) {
        for (int x : flo[b]) set_st(x, x);
        int xr = flo_from[b][g[b][pa[b]].u], xs = -1;
        for (int x : split_flo(flo[b], xr)) {
            if (xs == -1) { xs = x; continue; }
            pa[xs] = g[x][xs].u, S[xs] = 1, S[x] = 0;
            slk[xs] = 0, set_slk(x), q_push(x), xs = -1;
        } for (int x : flo[b])
            if (x == xr) S[x] = 1, pa[x] = pa[b];
            else S[x] = -1, set_slk(x);
        st[b] = 0;
    }
    bool on_found_edge(const edge &e) {
        if (int u = st[e.u], v = st[e.v]; S[v] == -1) {
            int nu = st[match[v]]; pa[v] = e.u; S[v] = 1;
            slk[v] = slk[nu] = S[nu] = 0; q_push(nu);
        } else if (S[v] == 0) {
            if (int o = lca(u, v)) add_blossom(u, o, v);
            else return augment(u, v), augment(v, u), true;
        } return false;
    }
    bool matching() {
        fill(ALL(S), -1), fill(ALL(slk), 0);
        q = queue<int>();
        REP(x, 1, nx) if (st[x] == x && !match[x])
            pa[x] = S[x] = 0, q_push(x);
        if (q.empty()) return false;
        for (;;) {
            while (SZ(q)) {
                int u = q.front(); q.pop();
                if (S[st[u]] == 1) continue;
                REP(v, 1, n)
                    if (g[u][v].w > 0 && st[u] != st[v]) {
                        if (E(g[u][v]) != 0)
                            update_slk(u, st[v], slk[st[v]]);
                        else if (on_found_edge(g[u][v]))
                            return true;
                    }
            }
            int d = INF;
            REP(b, n + 1, nx) if (st[b] == b && S[b] == 1)
                d = min(d, lab[b] / 2);
            REP(x, 1, nx)
                if (int s = slk[x]; st[x] == x &&
                    s && S[x] <= 0)
                    d = min(d, E(g[s][x]) / (S[x] + 2));
            REP(u, 1, n)
                if (S[st[u]] == 1) lab[u] += d;
                else if (S[st[u]] == 0) {
                    if (lab[u] <= d) return false;
                    lab[u] -= d;
                } REP(b, n + 1,
                    nx) if (st[b] == b && S[b] >= 0)
                        lab[b] += d * (2 - 4 * S[b]);
            REP(x, 1, nx)
                if (int s = slk[x]; st[x] == x &&
                    s && st[s] != x && E(g[s][x]) == 0)
                    if (on_found_edge(g[s][x])) return true;
            REP(b, n + 1, nx)
                if (st[b] == b && S[b] == 1 && lab[b] == 0)
                    expand_blossom(b);
        }
    }
    pair<ll, int> solve() {
        fill(ALL(match), 0);
        REP(u, 0, n) st[u] = u, flo[u].clear();
        int w_max = 0;
        REP(u, 1, n) REP(v, 1, n) {
            flo_from[u][v] = (u == v ? u : 0);
            w_max = max(w_max, g[u][v].w);
        } fill(ALL(lab), w_max);
        int n_matches = 0; ll tot_weight = 0;
        while (matching()) ++n_matches;
        REP(u, 1, n) if (match[u] && match[u] < u)

```

```

    tot_weight += g[u][match[u]].w;
    return {tot_weight, n_matches};
}
void add_edge(int u, int v, int w) {
    g[u][v].w = g[v][u].w = w;
}
};

```

Usage:
 WeightGraph g(n); // 1-based
 g.add_edge(u, v, w);
 auto [sum, cnt] = g.solve();
 // total weight, number of matched pairs

4.6 SW-mincut [6621f5]

Global min cut, Stoer-Wagner. $O(V^3)$, adjacency matrix, undirected. 0-base.

```

struct SW { // global min cut,  $O(V^3)$ 
#define REP for (int i = 0; i < n; ++i)
    static const int MXN = 514, INF = 2147483647;
    int vst[MXN], edge[MXN][MXN], wei[MXN];
    void init(int n) { REP fill_n(edge[i], n, 0); }
    void addEdge(int u, int v, int w) {
        edge[u][v] += w; edge[v][u] += w;
    }
    int search(int &s, int &t, int n) {
        fill_n(vst, n, 0), fill_n(wei, n, 0);
        s = t = -1;
        int mx, cur;
        for (int j = 0; j < n; ++j) {
            mx = -1, cur = 0;
            REP if (wei[i] > mx) cur = i, mx = wei[i];
            vst[cur] = 1, wei[cur] = -1;
            s = t; t = cur;
            REP if (!vst[i]) wei[i] += edge[cur][i];
        } return mx;
    }
    int solve(int n) {
        int res = INF;
        for (int x, y; n > 1; n--) {
            res = min(res, search(x, y, n));
            REP edge[i][x] = (edge[x][i] += edge[y][i]);
            REP {
                edge[y][i] = edge[n - 1][i];
                edge[i][y] = edge[i][n - 1];
            }
        } return res;
    }
} sw;

```

Usage:
 sw.init(n); // undirected, adj matrix
 sw.addEdge(u, v, w); // weights accumulate
 int cut = sw.solve(n); // global min cut

4.7 BoundedFlow*(Dinic*) [3281c3]

Flow with lower bounds; feasible circulation or bounded max flow. Size arrays $n+3$. 0-base.

```

struct BoundedFlow { // 0-base
    struct edge { int to, cap, flow, rev; };
    vector<edge> G[N];
    int n, s, t, dis[N], cur[N], cnt[N];
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n + 2; ++i) G[i].clear(),
            cnt[i] = 0;
    }
    void add_edge(int u, int v, int lcap, int rcap) {
        cnt[u] -= lcap, cnt[v] += lcap;
        G[u].pb({v, rcap, lcap, SZ(G[v])});
        G[v].pb({u, 0, 0, SZ(G[u]) - 1});
    }
    void add_edge(int u, int v, int cap) {
        G[u].pb({v, cap, 0, SZ(G[v])});
        G[v].pb({u, 0, 0, SZ(G[u]) - 1});
    }
    int dfs(int u, int cap) {
        if (u == t || !cap) return cap;
        for (int &i = cur[u]; i < SZ(G[u]); ++i) {
            edge &e = G[u][i];
            if (dis[e.to] == dis[u] + 1 && e.cap != e.flow) {
                int df = dfs(e.to, min(e.cap - e.flow, cap));
                if (df) {
                    e.flow += df, G[e.to][e.rev].flow -= df;
                    return df;
                }
            }
        }
        dis[u] = -1;
        return 0;
    }
    bool bfs() {
        fill_n(dis, n + 3, -1);
        queue<int> q; q.push(s), dis[s] = 0;
    }
};

```

```

    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (edge &e : G[u])
            if (!dis[e.to] && e.flow != e.cap)
                q.push(e.to), dis[e.to] = dis[u] + 1;
    } return dis[t] != -1;
}
int maxflow(int _s, int _t) {
    s = _s, t = _t;
    int flow = 0, df;
    while (bfs()) {
        fill_n(cur, n + 3, 0);
        while ((df = dfs(s, INF))) flow += df;
    } return flow;
}
bool solve() {
    int sum = 0;
    for (int i = 0; i < n; ++i)
        if (cnt[i] > 0)
            add_edge(n + 1, i, cnt[i]), sum += cnt[i];
        else if (cnt[i] < 0) add_edge(i, n + 2, -cnt[i]);
    if (sum != maxflow(n + 1, n + 2)) sum = -1;
    for (int i = 0; i < n; ++i)
        if (cnt[i] > 0)
            G[n + 1].pop_back(), G[i].pop_back();
        else if (cnt[i] < 0)
            G[i].pop_back(), G[n + 2].pop_back();
    return sum != -1;
}
int solve(int _s, int _t) {
    add_edge(_t, _s, INF);
    if (!solve()) return -1;
    int fake = SZ(G[_t]) - 1, rev = G[_t][fake].rev;
    int x = G[_t][fake].flow;
    G[_t][fake].cap = G[_t][fake].flow;
    G[_s][rev].cap = G[_s][rev].flow;
    x += maxflow(_s, _t);
    return G[_t].pop_back(), G[_s].pop_back(), x;
}
};

```

Usage:
 static BoundedFlow bf; bf.init(n);
 bf.add_edge(u, v, lo, hi); // lower bound lo
 bool feasible = bf.solve(); // circulation
 // or: int f = bf.solve(s, t); // bounded max flow

4.8 Gomory Hu tree* [e7elce]

All-pairs min cut as a tree ($n-1$ max-flow runs, Gusfield). Needs a MaxFlow named Dinic. 0-base.

```

MaxFlow Dinic;
int g[MAXN], gh_w[MAXN];
void GomoryHu(int n) { // 0-base
    fill_n(g, n, 0), fill_n(gh_w, n, 0);
    for (int i = 1; i < n; ++i) {
        Dinic.reset();
        int p = g[i], cut = Dinic.maxflow(i, p);
        gh_w[i] = cut;
        for (int j = i + 1; j < n; ++j)
            if (g[j] == p && ~Dinic.dis[j])
                g[j] = i;
        if (p && ~Dinic.dis[g[p]])
            g[i] = g[p], g[p] = i,
            gh_w[i] = gh_w[p], gh_w[p] = cut;
    }
}

```

Usage:
 Dinic.init(n); // build graph in Dinic first
 GomoryHu(n); // g[i] = parent, gh_w[i] = edge weight to g[i]
 // mincut(u,v) = min gh_w[x] on the tree path u-v

4.9 Minimum Cost Circulation* [2cc37c]

Min cost circulation via capacity scaling; cancels negative cycles. $O(VE \cdot E \log C)$. 0-base.

```

struct MinCostCirculation { // 0-base
    struct Edge
    { ll from, to, cap, fcap, flow, cost, rev;
    } *past[N];
    vector<Edge> G[N];
    ll dis[N], inq[N], n;
    void BellmanFord(int s) {
        fill_n(dis, n, INF), fill_n(inq, n, 0);
        queue<int> q;
        auto relax = [&](int u, ll d, Edge *e) {
            if (dis[u] > d) {
                dis[u] = d, past[u] = e;
                if (!inq[u]) inq[u] = 1, q.push(u);
            }
        }; relax(s, 0, 0);
        while (!q.empty()) {

```

```

    int u = q.front(); q.pop(), inq[u] = 0;
    for (auto &e : G[u])
        if (e.cap > e.flow)
            relax(e.to, dis[u] + e.cost, &e);
}
}
void try_edge(Edge &cur) {
    if (cur.cap > cur.flow) return ++cur.cap, void();
    BellmanFord(cur.to);
    if (dis[cur.from] + cur.cost < 0) {
        ++cur.flow, --G[cur.to][cur.rev].flow;
        for (int i = cur.from;
            past[i]; i = past[i] -> from) {
            auto &e = *past[i];
            ++e.flow, --G[e.to][e.rev].flow;
        }
    } ++cur.cap;
}
void solve(int mxlg) {
    for (int b = mxlg; b >= 0; --b) {
        for (int i = 0; i < n; ++i)
            for (auto &e : G[i])
                e.cap *= 2, e.flow *= 2;
        for (int i = 0; i < n; ++i)
            for (auto &e : G[i])
                if (e.fcap >> b & 1) try_edge(e);
    }
}
void init(int _n)
{ n = _n; for (int i = 0; i < n; ++i) G[i].clear(); }
void add_edge(int a, int b, int cap, int cost) {
    G[a].pb({
        a, b, 0, cap, 0, cost, SZ(G[b]) + (a == b)
    }, G[b].pb({b, a, 0, 0, 0, -cost, SZ(G[a]) - 1});
}
} mcmf; // O(VE * ElogC)
Usage:
mcmf.init(n); // negative costs allowed
mcmf.add_edge(u, v, cap, cost);
mcmf.solve(mxlg);
// mxlg = highest set bit of max capacity

```

4.10 Flow Models

- Maximum/Minimum flow with lower bound / Circulation problem
 - Construct supersource S , sink T .
 - Edge (x, y, l, u) : connect $x \rightarrow y$ with cap $u - l$.
 - $in(v)$:= sum of incoming lower bounds minus sum of outgoing ones.
 - $in(v) > 0$: connect $S \rightarrow v$ with cap $in(v)$; else $v \rightarrow T$ with cap $-in(v)$.
 - Max: connect $t \rightarrow s$ with cap ∞ (skip for circulation), f := maxflow $S \rightarrow T$. No solution if $f \neq \sum_{in(v)>0} in(v)$; else maxflow $s \rightarrow t$ is the answer.
 - Min: f := maxflow $S \rightarrow T$; connect $t \rightarrow s$ with cap ∞ , f' := flow $S \rightarrow T$. No solution if $f + f' \neq \sum_{in(v)>0} in(v)$; else f' .
- Each edge e has solution $l_e + f_e$, f_e its flow.
- Construct minimum vertex cover from maximum matching M on bipartite graph (X, Y)
 - Redirect: $y \rightarrow x$ if $(x, y) \in M$, else $x \rightarrow y$.
 - DFS from unmatched vertices in X .
 - Choose $x \in X$ iff unvisited; $y \in Y$ iff visited.
- Minimum cost cyclic flow
 - Construct supersource S , sink T .
 - Edge (x, y, c) : if $c > 0$ connect $x \rightarrow y$ with $(cost, cap) = (c, 1)$, else $y \rightarrow x$ with $(-c, 1)$.
 - For $c < 0$ edges, sum cost as K , then $d(y)++$, $d(x)--$.
 - $d(v) > 0$: connect $S \rightarrow v$ with $(0, d(v))$; $d(v) < 0$: $v \rightarrow T$ with $(0, -d(v))$.
 - Flow $S \rightarrow T$; answer is the flow cost $C + K$.
- Maximum density induced subgraph
 - Binary search the answer T ; let K be the sum of all weights.
 - Connect $s \rightarrow v, v \in G$, with cap K .
 - Edge (u, v, w) : connect both ways with cap w .
 - Connect $v \rightarrow t$ with cap $K + 2T - (\sum_{e \in E(v)} w(e)) - 2w(v)$.
 - T is valid if maxflow $f < K|V|$.
- Minimum weight edge cover
 - Copy each $v \in V$ as v' ; connect $u' \rightarrow v'$ with weight $w(u, v)$.
 - Connect $v \rightarrow v'$ with weight $2\mu(v)$, $\mu(v)$:= cheapest edge at v .
 - Min weight perfect matching on G' .
- Project selection problem
 - $p_v > 0$: edge (s, v) cap p_v ; else (v, t) cap $-p_v$.
 - Edge (u, v) cap w := cost of choosing u without v .
 - The min cut equals the max profit over subsets of projects.
- Dual of minimum cost maximum flow
 - Cap c_{uv} , flow f_{uv} , cost w_{uv} , required flow difference b_u .
 - If all w_{uv} are integers, an optimum with all p_u integral exists.

$$\begin{aligned}
 & \min \sum_{uv} w_{uv} f_{uv} & \min \sum_u b_u p_u + \sum_{uv} c_{uv} \max(0, p_v - p_u - w_{uv}) \\
 & -f_{uv} \geq -c_{uv} \Leftrightarrow & \\
 & \sum_v f_{vu} - \sum_v f_{uv} = -b_u & p_u \geq 0
 \end{aligned}$$

4.11 matching

- 最大匹配 + 最小邊覆蓋 = V
- 最大獨立集 + 最小點覆蓋 = V
- 最大匹配 = 最小點覆蓋
- 最短路徑覆蓋數 = $V - \text{最大匹配數}$

4.12 ISAP* [3f9cf8]

Max flow with the improved shortest-augmenting-path method. The template uses vertices $0 \dots n-1$ and reserves $n+1, n+2$ as source/sink.

```

struct Maxflow {
    static const int MAXV = 20010;
    static const int INF = 1000000;
    struct Edge { int v, c, r; };
    int s, t;
    vector<Edge> G[MAXV * 2];
    int iter[MAXV * 2], d[MAXV * 2], gap[MAXV * 2], tot;
    void init(int x) {
        tot = x + 2;
        s = x + 1, t = x + 2;
        for (int i = 0; i <= tot; i++)
            G[i].clear(), iter[i] = d[i] = gap[i] = 0;
    }
    void addEdge(int u, int v, int c) {
        G[u].push_back({v, c, SZ(G[v])});
        G[v].push_back({u, 0, SZ(G[u]) - 1});
    }
    void bfs() {
        fill(d, d + tot + 1, tot);
        queue<int> q;
        d[t] = 0, q.push(t);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (auto &e : G[u])
                if (G[e.v][e.r].c > 0 && d[e.v] == tot)
                    d[e.v] = d[u] + 1, q.push(e.v);
        }
        fill(gap, gap + tot + 1, 0);
        for (int i = 0; i <= tot; ++i) ++gap[d[i]];
    }
    int dfs(int p, int flow) {
        if (p == t) return flow;
        for (int &i = iter[p]; i < SZ(G[p]); i++) {
            Edge &e = G[p][i];
            if (e.c && d[p] == d[e.v] + 1) {
                int f = dfs(e.v, min(flow, e.c));
                if (f) {
                    e.c -= f;
                    G[e.v][e.r].c += f;
                    return f;
                }
            }
        }
        if (!--gap[d[p]]) d[s] = tot;
        else {
            d[p]++, iter[p] = 0, ++gap[d[p]];
        }
        return 0;
    }
    int solve() {
        int res = 0;
        fill(iter, iter + tot + 1, 0);
        bfs();
        for (; d[s] < tot; res += dfs(s, INF));
        return res;
    }
} flow;
Usage:
flow.init(n); // vertices
are 0..n-1; the template uses flow.s=n+1, flow.t=n+2
flow.addEdge(u, v, c);
int max_flow = flow.solve();

```

5 String

5.1 KMP [8309a4]

Pattern search, $O(|A| + |B|)$. Prints every 1-based occurrence; an empty pattern matches all $|A| + 1$ boundaries. `nxt` is the prefix-function table.

```

int n = s.size(), m = t.size(); vector<int> nxt(m);
if (!m) for (int i = 0; i <= n; ++i) cout << i + 1 << endl;
else {
    for (int i = 1, j = 0; i < m; i++) {
        while (j && t[i] != t[j]) j = nxt[j - 1];
        if (t[i] == t[j]) j++; nxt[i] = j;
    }
    for (int i = 0, j = 0; i < n; i++) {
        while (j && s[i] != t[j]) j = nxt[j - 1];
        if (s[i] == t[j]) j++;
        if (j == m) cout << i - m + 2 << endl,
            j = nxt[j - 1];
    }
}

```

```

    }
}
Usage:
string s = "ababa", t = "aba";
// run the listing: prints 1 and 3
// nxt[] is the prefix-function table

```

5.2 Z-value* [9d8afe]

```

z[i] = lcp(s, s[i..]) in O(n). z[0] is left unset.
int z[MAXN];
void make_z(const string &s) {
    for (int i = 1, l = 0, r = 0; i < SZ(s); ++i) {
        z[i] = max(0, min(r - i + 1, z[i - l]));
        while (i + z[i] < SZ(s) &&
            s[i + z[i]] == s[z[i]]) ++z[i];
        if (i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
    }
}
Usage:
make_z(s); // z[i] = lcp(s, s[i..])
// z[0] is NOT set; assign SZ(s) if needed

```

5.3 Manacher* [1ad8ef]

All palindromic radii in $O(n)$ by interleaving %. 0-base.

```

int z[MAXN]; // 0-base
/* center i: radius z[i * 2 + 1] / 2
   center i, i + 1: radius z[i * 2 + 2] / 2
   both aba, abba have radius 2 */
void Manacher(string tmp) {
    string s = "%"; int l = 0, r = 0;
    for (char c : tmp) s.pb(c), s.pb('%');
    for (int i = 0; i < SZ(s); ++i) {
        z[i] = r > i ? min(z[2 * l - i], r - i) : 1;
        while (i - z[i] >= 0 && i + z[i] < SZ(s) &&
            s[i - z[i]] == s[i + z[i]]) ++z[i];
        if (z[i] + i > r) r = z[i] + i, l = i;
    }
}
Usage:
Manacher(s);
// centre i      -> radius z[i*2+1]/2
// centre i,i+1  -> radius z[i*2+2]/2

```

5.4 SAIS* [6f26bc]

Suffix array in $O(n)$ plus Kasai height. Needs C++20; input must not contain a 0 byte.

```

auto sais(const auto &s) {
    const int n = SZ(s), z = ranges::max(s) + 1;
    if (n == 1) return vector{0};
    vector<int> c(z); for (int x : s) ++c[x];
    partial_sum(ALL(c), begin(c));
    vector<int> sa(n); auto I = views::iota(0, n);
    vector<bool> t(n, true);
    for (int i = n - 2; i >= 0; --i) t[i] = (s[i] ==
        s[i + 1] ? t[i + 1] : s[i] < s[i + 1]);
    auto is_lms = views::filter([&t](int x) {
        return x && t[x] && !t[x - 1]; });
    auto induce = [&] { for (auto x = c; int y :
        sa) if (y-- && (!t[y]) sa[x[s[y] - 1]++] = y;
        for (auto x = c; int y : sa | views::
            reverse) if (y-- && (t[y]) sa[--x[s[y]]] = y;
        };
    vector<int> lms, q(n); lms.reserve(n);
    for (auto x = c; int i : I | is_lms) q[i] = SZ(lms),
        lms.pb(sa[--x[s[i]]] = i);
    induce(); vector<int> ns(SZ(lms));
    for (int j = -1, nz = 0; int i : sa | is_lms) {
        if (j >= 0) {
            int len = min({n - i, n - j, lms[q[i] + 1] - i});
            ns[q[i]] =
                nz += lexicographical_compare(begin(s) + j,
                    begin(s) + j
                        + len, begin(s) + i, begin(s) + i + len);
            j = i;
        }
        fill(ALL(sa), 0); auto nsa = sais(ns);
        for (auto x = c; int y : nsa | views::
            reverse) y = lms[y], sa[--x[s[y]]] = y;
        return induce(), sa;
    }
}
// sa[i]: sa[i]-th suffix is the i-th lexicographically
// smallest suffix.
// hi[i]: LCP of suffix sa[i] and suffix sa[i - 1].
struct Suffix {
    int n; vector<int> sa, hi, ra;
    Suffix(const auto &s,
        int _n : n(_n), hi(n), ra(n) {
        vector<int> s(n + 1);
        copy_n(s, n, begin(s));
        // s[n] = 0; _s shouldn't contain 0
        sa = sais(s); sa.erase(sa.begin());
        for (int i = 0; i < n; ++i) ra[sa[i]] = i;
        for (int i = 0, h = 0; i < n; ++i) {
            if (!ra[i]) { h = 0;

```

```

                continue;
            } for (int j = sa[ra[i] - 1]; max(i,
                j) + h < n && s[i + h] == s[j + h];) ++h;
            hi[ra[i]] = h ? h - 1 : 0;
        }
    };
    Suffix sf(s.c_str(), s.size()); // no 0 bytes
    sf.sa[i]; // i-th smallest suffix
    sf.ra[i]; // rank of suffix i
    sf.hi[i]; // lcp(sa[i], sa[i-1]); needs C++20
}

```

5.5 Aho-Corasick Automatan* [36cbe5]

Multi-pattern matching. Build $O(\sum |s_i| \sigma)$, scan $O(|T|)$. Use rxn for $O(1)$ transitions; root is node 1, chars are c-'A'.

```

struct AC_Automatan {
    int nx[LEN][sigma], fl[LEN], cnt[LEN], ord[LEN], top;
    int rxn[LEN][sigma]; // node actually reached
    int newnode() {
        fill_n(nx[top], sigma, -1), fill_n(rxn[top],
            sigma, 0), fl[top] = cnt[top] = ord[top] = 0;
        return top++;
    }
    void init() { top = 1, newnode(); }
    int input(string &s) {
        int X = 1; for (char c : s) {
            if (!nx[X][c - 'A']) nx[X][c - 'A'] = newnode();
            X = nx[X][c - 'A'];
        } return X; // return the end node of string
    }
    void make_fl() {
        queue<int> q; q.push(1);
        for (int t = 0; !q.empty(); ) { int R = q.front();
            q.pop(), ord[t++] = R;
            for (int i = 0; i < sigma; ++i) if (~nx[R][i]) {
                int X = rxn[R][i] = nx[R][i], Z = fl[R];
                for (; Z && !~nx[Z][i]; ) Z = fl[Z];
                fl[X] = Z ? nx[Z][i] : 1, q.push(X);
                rxn[fl[R]][i] = R > 1 ?
                    rxn[fl[R]][i] : 1;
            }
        }
    }
    void solve() {
        for (int i = top - 2; i > 0;
            --i) cnt[fl[ord[i]]] += cnt[ord[i]];
    }
} ac;
// Usage:
ac.init();
int end = ac.input(pat); ac.cnt[end]++;
ac.make_fl(); // build fail + rxn
for (char c : text) x = ac.rxn[x][c - 'A'];
ac.solve(); // push cnt up the fail tree

```

5.6 Smallest Rotation [3164ef]

Lexicographically smallest rotation (Booth/min-notation) in $O(n)$.

```

string mcp(string s) {
    int n = SZ(s); if (n < 2) return s;
    int i = 0, j = 1; s += s;
    while (i < n && j < n) { int k = 0;
        while (k < n && s[i + k] == s[j + k]) ++k;
        if (s[i + k] <= s[j + k]) j += k + 1;
        else i += k + 1; if (i == j) ++j;
    } return s.substr(i < n ? i : j, n);
}

```

```

// Usage:
string t = mcp(s);
// lexicographically smallest rotation

```

5.7 De Bruijn sequence* [a09470]

Sequence of length $N+K-1$ whose length- N windows are all distinct ($K < C^N$).

```

constexpr int MAXC = 10, MAXN = 1e5 + 10;
struct DBSeq {
    int C, N, K, L, buf[MAXC * MAXN]; // K <= C^N
    void dfs(int *out, int t, int p, int &ptr) {
        if (ptr >= L) return; if (t > N) {
            if (N % p) return;
            for (int i = 1; i <= p &&
                ptr < L; ++i) out[ptr++] = buf[i];
        } else { buf[t] = buf[t - p], dfs(out, t + 1, p,
            ptr); for (int j = buf[t - p] + 1; j < C; ++j)
            buf[t] = j, dfs(out, t + 1, t, ptr);
        }
    }
    void solve(int _c, int _n, int _k, int *out) {
        int p = 0; C = _c, N = _n, K = _k, L = N + K - 1;
        dfs(out, 1, 1, p);
        if (p < L) fill(out + p, out + L, 0);
    }
} dbs;
// Usage:
dbs.solve(C, N, K, out);
// out has length N+K-1; every length-N
// window over it is distinct (K <= C^N)

```


5.8 Extended SAM* [64c3b7]

Generalised suffix automaton over many strings; cnt gives occurrences per state. Chars are 'a'.

```
struct exSAM {
    int len[N * 2], link[N * 2]; // maxlength, suflink
    int next[N * 2][CNUM], tot; // [0, tot), root = 0
    int lenSorted[N * 2]; // topo. order
    int cnt[N * 2]; // occurrence
    int newnode() {
        fill_n(next[tot], CNUM, 0);
        len[tot] = cnt[tot] = link[tot] = 0; return tot++;
    }
    void init() { tot = 0, newnode(), link[0] = -1; }
    int insertSAM(int last, int c) {
        int cur = next[last][c]; len[cur] = len[last] + 1;
        int p = link[last];
        while (p != -1 &&
            !next[p][c]) next[p][c] = cur, p = link[p];
        if (p == -1) return link[cur] = 0, cur;
        int q = next[p][c];
        if (len[p] + 1 == len[q]) return link[cur] = q, cur;
        int clone = newnode();
        for (int i = 0; i < CNUM; ++i) next[clone][i] =
            len[next[q][i]] ? next[q][i] : 0;
        len[clone] = len[p] + 1;
        while (p != -1 &&
            next[p][c] == q) next[p][c] = clone, p = link[p];
        link[link[cur] = clone] = link[q]; link[q] = clone;
        return cur;
    }
    void insert(const string &s) {
        int cur = 0; for (auto ch : s) {
            int &nxt = next[cur][int(ch - 'a')];
            if (!nxt) nxt = newnode(); cnt[cur = nxt] += 1;
        }
    }
    void build() {
        queue<int> q; q.push(0); while (!q.empty()) {
            int cur = q.front(); q.pop();
            for (int i = 0; i < CNUM; ++i) if (next[cur][i]) q.push(insertSAM(cur, i));
            vector<int> lc(tot);
            for (int i = 1; i < tot; ++i) ++lc[len[i]];
            partial_sum(ALL(lc), lc.begin());
            for (int i = 1; i < tot; ++i) lenSorted[-lc[len[i]]] = i;
        }
    }
    void solve() {
        for (int i = tot - 2; i >= 0; --i) cnt[link[lenSorted[i]]] += cnt[lenSorted[i]];
    }
};
Usage:
static exSAM ex; ex.init();
for (auto &s : strs) ex.insert(s); // trie
ex.build(); // trie -> generalised SAM
ex.solve(); // cnt = occurrences per state
```

5.9 PalTree* [529d56]

Eertree: counts distinct palindromic substrings and their occurrences in $O(n\sigma)$.

```
struct palindromic_tree {
    struct node {
        int next[26], fail, len;
        int cnt, num; // cnt: appear times, num: number of
        // pal. suf.
    };
    node(int l = 0) : fail(0), len(l), cnt(0), num(0) {
        fill(next, next + 26, 0);
    };
    vector<node> St;
    vector<char> s;
    int last, n;
    palindromic_tree() : St(2), last(1), n(0) {
        St[0].fail = 1, St[1].len = -1, s.pb(-1);
    }
    inline void clear() {
        St.clear(), s.clear(), last = 1, n = 0,
        St.pb(0), St.pb(-1), St[0].fail = 1, s.pb(-1);
    }
    inline int get_fail(int x) {
        while (s[n - St[x].len - 1] != s[n]) x = St[x].fail;
        return x;
    }
    inline void add(int c) {
        s.push_back(c - 'a'), ++n;
        int cur = get_fail(last);
        if (!St[cur].next[c]) {
            int now = SZ(St);
            St.pb(St[cur].len + 2);
            St[now].fail =

```

```
St[get_fail(St[cur].fail)].next[c],
St[cur].next[c] = now,
St[now].num = St[St[now].fail].num + 1;
} last = St[cur].next[c], ++St[last].cnt;
}
inline void count() { // counting cnt
    for (auto i = St.rbegin(); i !=
        St.rend(); ++i) St[i->fail].cnt += i->cnt;
}
inline int size() { // The number of diff. pal.
    return SZ(St) - 2;
}
};
Usage:
palindromic_tree pt;
for (char c : s) pt.add(c);
pt.count(); // fill cnt (occurrences)
pt.size(); // # distinct palindromes
```

5.10 Main Lorentz [567ef6]

All tandem repeats in $O(n \log n)$, stored as start-ranges per length. Handles every byte value by encoding an out-of-alphabet separator internally.

```
vector<pair<int, int>> rep[kN]; // 0-base [l, r]
void main_lorentz(const string &s, int sft = 0) {
    const int n = s.size(); if (n <= 1) return;
    const int nu = n / 2, nv = n - nu;
    const string u = s.substr(0, nu), v = s.substr(nu,
        ru(u.rbegin(), u.rend()), rv(v.rbegin(), v.rend()));
    main_lorentz(u, sft), main_lorentz(v, sft + nu);
    auto join = [](const string &a, const string &b) {
        vector<int> x; x.reserve(a.size() + b.size() + 1);
        for (unsigned char c : a) x.push_back(c);
        x.push_back(256);
        for (unsigned char c : b) x.push_back(c); return x;
    };
    auto zalgo = [](const vector<int> &a) {
        vector<int> z(a.size());
        for (int i = 1, l = 0,
            r = 0; i < (int)a.size(); ++i) {
            if (i <= r) z[i] = min(r - i + 1, z[i - 1]);
            while (i + z[i] < (int)a.size() &&
                a[z[i]] == a[i + z[i]]) ++z[i];
            if (i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
        } return z;
    };
    const auto z1 = zalgo(join(ru, "")),
        z2 = zalgo(join(v, u)), z3 = zalgo(join(ru,
        rv)), z4 = zalgo(join(v, ""));
    auto get_z = [](const vector<int> &z, int i) {
        return (0 <= i and i < (int)z.size()) ? z[i] : 0;
    };
    auto add_rep = [&](bool left,
        int c, int l, int k1, int k2) {
        const int L = max(1, l - k2),
            R = min(l - left, k1);
        if (L > R) return; int d = !left * (l - 1);
        rep[l].emplace_back(sft + c - R - d,
            sft + c - L - d);
    };
    for (int cntr = 0; cntr < n; cntr++) {
        int l, k1, k2;
        if (cntr < nu) {
            l = nu - cntr, k1 = get_z(z1, nu - cntr),
            k2 = get_z(z2, nv + 1 + cntr);
        } else {
            l = cntr - nu + 1, k1 = get_z(z3, n + nu -
            cntr), k2 = get_z(z4, cntr - nu + 1);
        } if (k1 + k2 >= 1) add_rep(cntr < nu,
            cntr, l, k1, k2);
    }
} // p \in [l, r] => s[p, p + i] = s[p + i, p + 2i]
Usage:
main_lorentz(s); // fills rep[length]
// [L,R] in rep[l]: every p has s[p,p+1] == s[p+1,p+2l]
```

6 Math

6.1 ax+by=gcd(only exgcd *) [5bc171]

Extended Euclid, $O(\log \min(a, b))$. The trailing comment shows how to shift to the smallest non-negative solution.

```
p11 exgcd(ll a, ll b) {
    if (b == 0) return p11(1, 0); p11 q = exgcd
        (b, a % b); return p11(q.Y, q.X - q.Y * (a / b));
}
/* ax+by=res, let x be minimum non-negative
g, p = gcd(a, b), exgcd(a, b) * res / g
if p.X < 0: t = (abs(p.X) + b / g - 1) / (b / g)
else: t = -(p.X / (b / g))
p += (b / g, -a / g) * t */
```


6.2 Floor and Ceil [692c04]

Floor/ceil division that is correct for **negative** operands ($C++$ truncates toward zero).

```
int floor(int a, int b) {
    return a / b - (a % b && (a < 0) ^ (b < 0));
}
int ceil(int a, int b) {
    return a / b + (a % b && (a < 0) ^ (b > 0));
}
```

6.3 Floor Enumeration [7cbcdf]

Enumerates the $O(\sqrt{n})$ distinct values of $\lfloor n/i \rfloor$ with their index ranges.

```
// enumerating x = floor(n / i), [l, r]
for (int l = 1, r; l <= n; l = r + 1) { int x = n / l;
    r = n / x; }
```

6.4 Mod Min [899d01]

Smallest k with $l \leq k \leq m \leq r$, $O(\log m)$ by a Euclid-like recursion; -1 if none.

```
// min{k | l <= ((ak) mod m) <= r}, no solution -> -1
ll mod_min(ll a, ll m, ll l, ll r) {
    if (a == 0) return l ? -1 : 0;
    if (ll k = (l + a - 1) / a; k * a <= r) return k;
    ll b = m / a, c = m % a;
    if (ll y = mod_min(c, a, a - r % a, a - l % a); y != -1)
        return (l + y * c + a - 1) / a + y * b;
    return -1;
}
```

6.5 Linear Mod Inverse [232adb]

All inverses $1..N$ in $O(N)$. Requires a prime modulus and $N < \text{mod}$.

```
inv[1] = 1;
for (int i = 2; i <= N; ++i) inv[i] = 1LL *
    (mod - mod / i) * inv[mod % i] % mod;
```

6.6 Linear Filter Mu [ac2ac3]

Mobius function by linear sieve, $O(n)$; also leaves the prime list in p .

```
void getMu() {
    mu[1] = 1; for (int i = 2; i <= n; ++i) {
        if (!flg[i]) p[++tot] = i, mu[i] = -1;
        for (int j = 1; j <= tot && i * p[j] <= n; ++j) {
            flg[i * p[j]] = 1;
            if (i % p[j] == 0) { mu[i * p[j]] = 0;
                break; } mu[i * p[j]] = -mu[i];
        }
    }
}
```

6.7 Gaussian integer gcd [abafd0]

GCD over $\mathbb{Z}[i]$; the `rnd` macro rounds the quotient to the nearest Gaussian integer.

```
cpx gaussian_gcd(cpx a, cpx b) {
#define rnd
    (a, b) ((a >= 0 ? a * 2 + b : a * 2 - b) / (b * 2))
    ll c = a.real() * b.real() + a.imag() * b.imag(),
        d = a.imag() * b.real() - a.real() * b.imag(),
        r = b.real() * b.real() + b.imag() * b.imag();
    if (c % r == 0 && d % r == 0) return b;
    return gaussian_gcd(b,
        a - cpx(rnd(c, r), rnd(d, r)) * b);
}
```

6.8 Gauss Elimination [87881e]

Gauss-Jordan on doubles, $O(n^3)$. Uses a small pivot epsilon and leaves free variables at zero; use the fraction version when exactness matters.

```
void GAS(V<V<double>>&vc) {
    int len = vc.size(), row = 0;
    for (int col = 0; col < len && row < len; ++col) {
        int idx = row;
        while (idx < len &&
            fabs(vc[idx][col]) < 1e-12) ++idx;
        if (idx == len) continue;
        swap(vc[idx], vc[row]);
        double pivot = vc[row][col];
        for (double &x : vc[row]) x /= pivot;
        for (int j = 0; j < len; ++j) { if (j == row ||
            fabs(vc[j][col]) < 1e-12) continue;
            double mul = vc[j][col];
            for (int k = 0; k < (int)vc[j].size();
                ++k) vc[j][k] -= mul * vc[row][k];
            }
        ++row;
    }
}
```

6.9 floor sum* [dc29f6]

$\sum_{i=0}^{n-1} \lfloor (ai+b)/m \rfloor$ in $O(\log)$ by a Euclid-like recursion.

```
ll floor_sum(ll n, ll m, ll a, ll b) {
    ll ans = 0;
    if (a >= m) ans += (n - 1) * n * (a / m) / 2, a %= m;
    if (b >= m) ans += n * (b / m), b %= m;
    ll y_max = (a * n + b) / m, x_max = (y_max * m - b);
    if (y_max == 0) return ans;
    ans += (n - (x_max + a - 1) / a) * y_max;
    return ans + floor_sum(y_max,
        a, m, (a - x_max % a) % a);
}
// sum_{i=0}^{n-1} floor((a * i + b) / m) in log(n + m + a + b)
```

6.10 Miller Rabin* [9f7dec]

Deterministic primality with the witness sets listed in the file (7 bases below 2^{64}). Needs a 128-bit mul().

```
// n < 4,759,123,141      3 : 2, 7, 61
// n < 1,122,004,669,633  4 : 2, 13, 23, 1662803
// n < 3,474,749,660,383  6 : primes <= 13
// n < 2^64               7 :
// 2, 325, 9375, 28178, 450775, 9780504, 1795265022
bool Miller_Rabin(ll a, ll n) {
    if ((a = a % n) == 0) return 1;
    if (n % 2 == 0) return n == 2;
    ll tmp = n - 1,
        t = countr_zero((unsigned long long)tmp), x = 1;
    tmp >>= t;
    for (; tmp; tmp >>= 1, a = mul(a, a, n))
        if (tmp & 1) x = mul(x, a, n);
    if (x == 1 || x == n - 1) return 1;
    while (--t) if ((x = mul(x, x, n)) == n - 1) return 1;
    return 0;
}
```

Usage:

```
bool prime(ll n) {
    if (n < 2) return false;
    for (ll a : {2,325,9375,28178,450775,9780504,1795265022})
        if (!Miller_Rabin(a, n)) return false;
    return true;
}
```

6.11 Fraction [4ab37a]

Exact rational with auto-reduction. ll **numerator/denominator overflow easily** after a few operations; no divide-by-zero check.

```
struct fraction {
    ll n, d;
    fraction(const ll &n=0, const ll &d=1): n(_n),
        d(_d) { ll t = gcd(n, d);
        n /= t, d /= t; if (d < 0) n = -n, d = -d; }
    fraction operator-() const { return fraction(-n, d); }
    fraction operator+(const fraction &b) const {
        return fraction(n * b.d + b.n * d, d * b.d); }
    fraction operator-(const fraction &b) const {
        return fraction(n * b.d - b.n * d, d * b.d); }
    fraction operator*(const fraction &b) const {
        return fraction(n * b.n, d * b.d); }
    fraction operator/(const fraction &b) const {
        return fraction(n * b.d, d * b.n); }
    void print() { cout << n;
        if (d != 1) cout << "/" << d; }
};
```

6.12 Simultaneous Equations [c999bb]

Exact Gauss-Jordan over fraction; returns -1 if inconsistent, else the rank. Requires `Fraction.cpp`.

```
struct matrix { //m variables, n equations
    int n, m;
    fraction M[MAXN][MAXN + 1], sol[MAXN];
    int solve() { // -1: inconsistent, >= 0: rank
        int rank = 0;
        for (int col = 0; col < m && rank < n; ++col) {
            int row = rank;
            while (row < n && !M[row][col].n) ++row;
            if (row == n) continue;
            for (int k = 0; k <= m; ++k) swap(M[rank][k],
                M[row][k]);
            for (int j = 0; j < n; ++j) {
                if (rank == j) continue;
                fraction tmp = -M[j][col] / M[rank][col];
                for (int k = 0; k <=
                    m; ++k) M[j][k] = tmp * M[rank][k] + M[j][k];
            }
            ++rank;
        }
        for (int i = 0; i < n; ++i) { int piv = 0;
```

```

    while (piv < m && !M[i][piv].n) ++piv;
    if (piv == m && M[i][m].n) return -1; }
fill_n(sol, m, fraction());
for (int i = 0; i < n; ++i) { int piv = 0;
while (piv < m && !M[i][piv].n) ++piv;
if (piv < m) sol[piv] = M[i][m] / M[i][piv]; }
return rank; }
};
Usage:
matrix eq{}; eq.n = eq.m = 2;
eq.M[0][0]=1; eq.M[0][1]=1; eq.M[0][2]=3;
eq.M[1][0]=2; eq.M[1][1]=-1; eq.M[1][2]=0;
int rank=eq.solve(); // sol={1,2}; -1: inconsistent

```

6.13 Pollard Rho* [fdef9b]

Factorises in about $O(n^{1/4})$; results accumulate in cnt. Needs prime() and a 128-bit mul().

```

map<ll, int> cnt;
void PollardRho(ll n) {
    if (n == 1) return;
    if (prime(n)) return ++cnt[n], void();
    if (n % 2 == 0) return PollardRho(n / 2),
        ++cnt[2], void();
    ll x = 2, y = 2, d = 1, p = 1;
    #define f(x, n, p) ((mul(x, x, n) + p) % n)
    while (true) {
        if (d != n && d != 1) { PollardRho(n / d);
            PollardRho(d); return; }
        if (d == n) ++p;
        x = f(x, n, p), y = f(f(y, n, p), n, p);
        d = gcd(abs(x - y), n);
    }
}
Usage:
cnt.clear();
PollardRho(n);
for (auto [p, e] : cnt) // n = product(p^e)
    use_factor(p, e);

```

6.14 Simplex Algorithm [1d0b36]

max cx s.t. $Ax \leq b, x \geq 0$; -1 means unbounded or infeasible. n is the constraint count, m the variable count, and all arrays are 0-indexed.

```

const int MAXN = 11000, MAXM = 405;
const double eps = 1E-10;
double a[MAXN][MAXM], b[MAXN], c[MAXN];
double d[MAXN][MAXM], x[MAXN];
int ix[MAXN + MAXM]; // !!! array all indexed from 0
// max{cx} s.t. {Ax<=b,x>=0}; n: constraints, m: vars
// !!!
// x[] is the optimal solution vector
// usage: value = simplex(n, m);
double simplex(int n, int m) {
    ++m; fill_n(d[n], m + 1, 0);
    fill_n(d[n + 1], m + 1, 0);
    iota(ix, ix + n + m, 0); int r = n, s = m - 1;
    for (int i = 0; i < n; ++i) {
        for (int j = 0; j < m - 1; ++j) d[i][j] = -a[i][j];
        d[i][m - 1] = 1; d[i][m] = b[i];
        if (d[r][m] > d[i][m]) r = i;
    } copy_n(c, m - 1, d[n]);
    d[n + 1][m - 1] = -1;
    for (double dd;; ) {
        if (r < n) {
            swap(ix[s], ix[r + m]);
            d[r][s] = 1.0 / d[r][s];
            for (int j = 0; j <= m; ++j) if (j != s) d[r][j] *= -d[r][s];
            for (int i = 0; i <= n + 1; ++i) if (i != r) {
                for (int j = 0; j <= m; ++j) if (j != s)
                    d[i][j] += d[r][j] * d[i][s];
                d[i][s] *= d[r][s];
            } r = s = -1;
        }
        for (int j = 0; j < m; ++j) if (s < 0 || ix[s] > ix[j]) if (d[n + 1][j] > eps || (d[n + 1][j] > -eps && d[n][j] > eps)) s = j;
        if (s < 0) break;
        for (int i = 0; i < n; ++i) if (d[i][s] < -eps) if (r < 0 || (dd = d[r][m] / d[r][s] - d[i][m] / d[i][s]) < -eps || (dd < eps && ix[r + m] > ix[i + m])) r = i;
        if (r < 0) return -1; // not bounded
    } if (d[n + 1][m] < -eps) return -1;
    // not executable
    double ans = 0;
    fill_n(x, m, 0);
    // the missing enumerated x[i] = 0
    for (int i = m; i < n + m; ++i) if (ix[i] < m - 1) ans += d[i - m][m] * c[ix[i]], x[ix[i]] = d[i - m][m];

```

```

    return ans;
}
Usage:
int n=3, m=2; // max 3x+2y: x+y<=4, x<=2, y<=3
a[0][0]=1; a[0][1]=1; b[0]=4;
a[1][0]=1; b[1]=2; a[2][1]=1; b[2]=3;
c[0]=3; c[1]=2;
double value=simplex(n,m); // value=10, x={2,2}

```

6.15 chineseRemainder [0e3286]

CRT for non-coprime moduli; -1 if no solution. **Overflows ll easily** — use int128 for large moduli. Iterate for more than two congruences.

```

ll solve(ll x1, ll m1, ll x2, ll m2) {
    ll g = gcd(m1, m2); __int128 d = (__int128)x2 - x1;
    if (d % g) return -1; // no sol
    m1 /= g, m2 /= g; p11 p = exgcd(m1, m2);
    __int128 t = (__int128)p.first * (d / g) % m2;
    if (t < 0) t += m2;
    __int128 lcm = (__int128)m1 * m2 * g;
    __int128 res = (__int128)x1 + (__int128)m1 * g * t;
    return (ll)((res % lcm + lcm) % lcm);
}
Usage:
ll x = solve(2, 6, 5, 9); // x=14 (mod lcm=18)

```

6.16 Factorial without prime factor* [995ea7]

$n!$ with all factors p removed, mod p^k , in $O(p^k + \log^2 n)$. Core of Lucas for prime-power moduli.

```

// O(p^k + log^2 n), pk = p^k
ll prod[MAXP];
ll fac_no_p(ll n, ll p, ll pk) {
    prod[0] = 1; for (int i = 1; i <= pk; ++i)
        if (i % p) prod[i] = prod[i - 1] * i % pk;
        else prod[i] = prod[i - 1];
    ll rt = 1; for (; n; n /= p) {
        rt = rt * mpow(prod[pk], n / pk,
            pk) % pk, rt = rt * prod[n % pk] % pk;
    } return rt;
} // (n! without factor p) % p^k
Usage:
ll value=fac_no_p(10,3,27);
// value=7: remove every factor 3 from 10!, then mod 27

```

6.17 PiCount* [cad6d4]

$\pi(n)$ in about $O(n^{2/3})$; handles $n \sim 10^{13}$ in under 2s.

```

ll PrimeCount(ll n) { // n ~ 10^13 => < 2s
    if (n <= 1) return 0;
    int v = sqrt(n), s = (v + 1) / 2, pc = 0;
    vector<int> smalls(v + 1), skip(v + 1), roughs(s);
    vector<ll> larges(s);
    for (int i = 2; i <= v; ++i) smalls[i] = (i + 1) / 2;
    for (int i = 0; i < s; ++i) { roughs[i] = 2 * i + 1;
        larges[i] = (n / (2 * i + 1) + 1) / 2;
    } for (int p = 3; p <= v; ++p) {
        if (smalls[p] > smalls[p - 1]) {
            int q = p * p; ++pc;
            if (1LL * q * q > n) break;
            skip[p] = 1;
            for (int i = q; i <= v; i += 2 * p) skip[i] = 1;
            int ns = 0; for (int k = 0; k < s; ++k) {
                int i = roughs[k]; if (skip[i]) continue;
                ll d = 1LL * i * p;
                larges[ns] = larges[k] - (d <= v ?
                    larges[smalls[d] - pc] : smalls[n / d]) + pc;
                roughs[ns++] = i;
            } s = ns;
            for (int j = v / p; j >= p; --j) {
                int c = smalls[j] - pc, e = min(j * p + p, v + 1);
                for (int i = j * p; i < e; ++i) smalls[i] -= c;
            }
        }
    } for (int k = 1; k < s; ++k) {
        const ll m = n / roughs[k];
        ll t = larges[k] - (pc + k - 1);
        for (int l = 1; l < k; ++l) {
            int p = roughs[l]; if (1LL * p * p > m) break;
            t -= smalls[m / p] - (pc + l - 1);
        } larges[0] -= t;
    } return larges[0];
}

```

6.18 Discrete Log* [7a4e6c]

Smallest k with $x^k \equiv y \pmod{m}$ in $O(\sqrt{m})$. Does not require $\gcd(x, m) = 1$; verifies the answer before returning.

```

int DiscreteLog(int s, int x, int y, int m) {
    constexpr int kStep = 32000;
    unordered_map<int, int> p;
    int b = 1; for (int i = 0; i < kStep; ++i) {
        p[y] = i, y = 1LL * y * x % m, b = 1LL * b * x % m;
    } for (int i = 0; i < m + 10; i += kStep) {

```

```

    s = 1LL * s * b % m;
    if (p.find(s) != p.end()) return i + kStep - p[s];
} return -1;
}
int DiscreteLog(int x, int y, int m) {
    if (m == 1) return 0; int s = 1;
    for (int i = 0; i < 100; ++i) { if (s == y) return i;
        s = 1LL * s * x % m; } if (s == y) return 100;
    int p = 100 + DiscreteLog(s, x, y, m);
    if (fpow(x, p, m) != y) return -1;
    return p;
}

```

Usage:
int k = DiscreteLog(2, 8, 13); // k=3; -1 if none

6.19 Berlekamp Massey [3eb6fa]

Shortest linear recurrence of a sequence in $O(n^2)$; feed the result to LinearRecursion for the n -th term.

```

template <typename T>
vector<T> BerlekampMassey(const vector<T> &output) {
    vector<T> d(SZ(output) + 1), me, he;
    for (int f = 0, i = 1; i <= SZ(output); ++i) {
        for (int j = 0; j < SZ(me); ++j) d[i] += output[i - j - 2] * me[j];
        if ((d[i] - output[i - 1]) == 0) continue;
        if (me.empty()) { me.resize(f = i);
            continue; } vector<T> o(i - f - 1);
        T k = -d[i] / d[f]; o.pb(-k);
        for (T x : he) o.pb(x * k);
        o.resize(max(SZ(o), SZ(me)));
        for (int j = 0; j < SZ(me); ++j) o[j] += me[j];
        if (i - f + SZ(he) >= SZ(me)) he = me, f = i;
        me = o;
    } return me;
}

```

Usage:
vector<mint> seq = {0,1,1,2,3,5,8};
auto coef = BerlekampMassey(seq); // {1,1}

6.20 Primes

Handy primes for hashing and NTT moduli, from small values up to $2^{61} - 1$ and near 2^{64} .

```

/* 12721 13331 14341 75577 123457 222557 556679 999983
98789101 100102021 987654361 987777733 999888733
999991231 999991921 999997771 1000512343 1001010013
1010101333 1010102101 1076767633 1097774749
1000000000039 1000000000000037 2305843009213693951
4611686018427387847 9223372036854775783
18446744073709551557 */

```

6.21 Theorem

$$\sum_{d=1}^n \mu(d) \lfloor n/d \rfloor = 1 = \sum_{k=1}^n \sum_{d|k} \mu(d). \quad n = \sum_{d|n} \mu(d) \sigma(n/d) = \sum_{d|n} \mu(n/d) \sigma(d). \quad \sum_{k=1}^n \sigma(k) = \sum_{d=1}^n d \lfloor n/d \rfloor. \quad \varphi(n) = \sum_{d|n} \mu(d) n/d.$$

Lucas's Theorem

For non-negative integer n, m and prime P ,
 $C(m, n) \bmod P = C(m/M, n/M) * C(m \% M, n \% M) \bmod P$
 $= \text{mult_i}(C(m_i, n_i))$

where m_i is the i -th digit of m in base P .

Kirchhoff's theorem

$A_{\{ii\}} = \deg(i), A_{\{ij\}} = (i, j) \text{ in } E? - 1 : 0$

Deleting any one row, one column, and call the $\det(A)$

Nth Catalan recursive function:

$C_0 = 1, C_{n+1} = C_n * 2(2n+1)/(n+2)$

Mobius Formula

$u(n) = 1$, if $n = 1$

$(-1)^m$, 若 n 無平方數因數, 且 $n = p_1 * p_2 * p_3 * \dots * p_k$

0, 若 n 有大於 1 的平方數因數

- Property

1. (積性函數) $u(a)u(b) = u(ab)$

2. $\sum_{d|n} u(d) = [n=1]$

Mobius Inversion Formula

if $f(n) = \sum_{d|n} g(d)$
 then $g(n) = \sum_{d|n} u(d/n) f(d)$
 $= \sum_{d|n} u(d) f(n/d)$

- Application

the number/power of $\gcd(i, j) = k$

- Trick

分塊, $O(\sqrt{n})$

Chinese Remainder Theorem (m_i 兩兩互質)

$x = a_1 \pmod{m_1}$

$x = a_2 \pmod{m_2}$

....

$x = a_i \pmod{m_i}$

construct a solution:

Let $M = m_1 * m_2 * m_3 * \dots * m_n$

Let $M_i = M / m_i$

$t_i = 1 / M_i$
 $t_i * M_i = 1 \pmod{m_i}$
 solution $x = a_1 * t_1 * M_1 + a_2 * t_2 * M_2 + \dots + a_n * t_n * M_n + k * M$
 $= k * M + \text{suma_i} * t_i * M_i$, k is positive integer.
 under mod M , there is one solution $x = \text{suma_i} * t_i * M_i$
 Burnside's lemma
 $|G| * |X/G| = \sum (|X^g|)$ where g in G
 總方法數: 每一種旋轉下不動點的個數總和除以旋轉的方法數

6.22 Estimation

n | 2 3 4 5 6 7 8 9 20 30 40 50 100

$p(n)$ | 2 3 5 7 11 15 22 30 62 75 60 44 42 52 2e8

n | 100 1e3 1e6 1e9 1e12 1e15 1e18

$d(i)$ | 12 32 240 1344 6720 26880 103680

n | 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

$\binom{2n}{n}$ | 2 6 20 70 252 924 3432 12870 48620 184756 7e5 2e6 1e7 4e7 1.5e8

n | 2 3 4 5 6 7 8 9 10 11 12 13

B_n | 2 5 15 52 203 877 4140 21147 115975 7e5 4e6 3e7

6.23 General Purpose Numbers

• Bernoulli numbers

$B_0 = 1, B_1^{\pm} = \pm \frac{1}{2}, B_2 = \frac{1}{6}, B_3 = 0$

$\sum_{j=0}^m \binom{m+1}{j} B_j = 0$, EGF is $B(x) = \frac{x}{e^x - 1} = \sum_{n=0}^{\infty} B_n \frac{x^n}{n!}$.

$S_m(n) = \sum_{k=1}^n k^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k^+ n^{m+1-k}$

• Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$S(n, k) = S(n-1, k-1) + k S(n-1, k), S(n, 1) = S(n, n) = 1$

$S(n, k) = \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n$

$x^n = \sum_{i=0}^n S(n, i) (x)_i$

• Pentagonal number theorem

$\prod_{k=1}^{\infty} (1 - x^k) = 1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{k(3k+1)/2} + x^{k(3k-1)/2} \right)$

• Catalan numbers

$C_n^{(k)} = \frac{1}{(k-1)n+1} \binom{kn}{n}, C^{(k)}(x) = 1 + x[C^{(k)}(x)]^k$

• Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j: s.s.t. $\pi(j) > \pi(j+1), k+1$ j: s.s.t. $\pi(j) \geq j$, k j: s.s.t. $\pi(j) > j$.

$E(n, k) = (n-k)E(n-1, k-1) + (k+1)E(n-1, k)$

$E(n, 0) = E(n, n-1) = 1$

$E(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$

6.24 Tips for Generating Functions

• Ordinary Generating Function $A(x) = \sum_{i \geq 0} a_i x^i$

$- A(rx) \Rightarrow r^n a_n$

$- A(x) + B(x) \Rightarrow a_n + b_n$

$- A(x)B(x) \Rightarrow \sum_{i=0}^n a_i b_{n-i}$

$- A(x)^k \Rightarrow \sum_{i_1+i_2+\dots+i_k=n} a_{i_1} a_{i_2} \dots a_{i_k}$

$- xA(x)^k \Rightarrow na_n$

$- \frac{A(x)}{1-x} \Rightarrow \sum_{i=0}^n a_i$

• Exponential Generating Function $A(x) = \sum_{i \geq 0} \frac{a_i}{i!} x^i$

$- A(x) + B(x) \Rightarrow a_n + b_n$

$- A^{(k)}(x) \Rightarrow a_{n+k}$

$- A(x)B(x) \Rightarrow \sum_{i=0}^n \binom{n}{i} a_i b_{n-i}$

$- A(x)^k \Rightarrow \sum_{i_1+i_2+\dots+i_k=n} \binom{n}{i_1, i_2, \dots, i_k} a_{i_1} a_{i_2} \dots a_{i_k}$

$- xA(x) \Rightarrow na_n$

• Special Generating Function

$- (1+x)^n = \sum_{i \geq 0} \binom{n}{i} x^i$

$- \frac{1}{(1-x)^n} = \sum_{i \geq 0} \binom{n-1}{i} x^i$

7 Polynomial

7.1 Fast Fourier Transform [f3ed03]

Iterative complex FFT and integer convolution via mult, $O(n \log n)$. Returns exactly $|a| + |b| - 1$ rounded coefficients; double precision limits coefficient size.

using cd = complex<double>;

const double pi = acos(-1);

void fft(vector<cd> &vec, **bool** inv=false) {

int n = vec.size();

for(int i=0, j=0; i<n; ++i) { // j = 反過來模擬加法

if(i<j) swap(vec[i], vec[j]);

for(int bit=n>>1; (j^=bit)<bit; bit>>=1);

}

for(int len=2; len<=n; len<<=1) { // 循環長度 = 2pi

double ang = 2*pi/len*(inv?-1:1);

cd wlen(cos(ang), sin(ang));

for(int i=0; i<n; i+=len) {

cd w=1;

for(int j=0; j<len/2; ++j) {

// i = 定範圍, j = 去跑 merge

cd a = vec[i+j], b = w*vec[i+len/2+j];

vec[i+j] = a+b;

```

        vec[i+len/2+j] = a-b;
        w *= wlen;
    }
}
if(inv) for(auto &i : vec) i/=n;
}
#define ifft(vec) fft(vec,true)

vector<int> mult(const vector<int>& a,
               const vector<int>& b) {
    int sz = a.size()+b.size()-1, n=1; while(n<sz) n<=<=1;
    vector<cd> ca(a.begin(),
                 a.end()), cb(b.begin(),b.end());
    ca.resize(n), cb.resize(n), fft(ca), fft(cb);
    for(int i=0; i<n; ++i) ca[i] *= cb[i];
    ifft(ca);
    vector<int> res(sz);
    for(int i=0; i<sz; ++i) res[i] = round(ca[i].real());
    return res;
}
Usage:
vector<int> c = mult(a, b); // exact output length
// rounded from double; keep coefficients moderate

```

Prime	Root	Prime	Root
7681	17	167772161	3
12289	11	104857601	3
40961	3	985661441	3
65537	3	998244353	3
786433	10	1107296257	10
5767169	3	2013265921	31
7340033	3	2810183681	11
23068673	3	2885681153	3
469762049	3	605028353	3

7.2 Number Theory Transform* [32acbd]

Iterative NTT and convolution via mult, $O(n \log n)$ modulo 998244353. The listing defines int as long long; contain or undefine the macro after use.

```

#define int long long
const int mod=998244353, g=3;

int pwr(int a, int b, int ret=1) {
    do ret = b&1?ret*a%mod:ret, a=a*a%mod;
    while(b>>=1); return ret;
}

void ntt(vector<int> &vec, bool inv=false) {
    int n = vec.size();
    for(int i=0, j=0; i<n; ++i) { // j = 反過來模擬加法
        if(i<j) swap(vec[i], vec[j]);
        for(int bit=n>>1; (j^=bit)<bit; bit>>=1);
    }
    for(int len=2; len<=n; len<=<=1) { // 循環長度 = mod-1
        int wlen = pwr(g, (mod-1)/len);
        if(inv) wlen = pwr(wlen, mod-2);
        for(int i=0; i<n; i+=len) {
            int w=1;
            for(int j=0; j<len/2; ++j) {
                // i = 定範圍, j = 去跑 merge
                int a = vec[i+j], b = w*vec[i+len/2+j]%mod;
                vec[i+j] = a+b<mod ? a+b : a+b-mod;
                vec[i+len/2+j] = a-b>=0 ? a-b : a-b+mod;
                w = w*wlen%mod;
            }
        }
    }
    if(inv) {
        int invn = pwr(n, mod-2);
        for(auto &i : vec) i = i*invn%mod;
    }
}
#define intt(vec) ntt(vec,true)

vector<int> mult(vector<int> a, vector<int> b) {
    int sz = a.size()+b.size()-1, n=1; while(n<sz) n<=<=1;
    a.resize(n), b.resize(n), ntt(a), ntt(b);
    for(int i=0; i<n; ++i) a[i] = a[i]*b[i] % mod;
    intt(a), a.resize(sz); return a;
}
Usage:
vector<int> c = mult(a, b); // modulo 998244353
#undef int // if the listing's macro must not leak

```

7.3 Fast Walsh Transform* [325770]

or/and/xorconvolution, $O(n \log n)$; file implements or. Also subset convolution in $O(2^L L^2)$.

```

/* x: a[j], y: a[j + (L >> 1)]
or: (y += x * op), and: (x += y * op)

```

```

xor: (x, y = (x + y) * op, (x - y) * op)
invop: or, and, xor = -1, -1, 1/2 */
void fwt(int *a, int n, int op) { //or
    for (int L = 2; L <= n; L <=<= 1)
        for (int i = 0; i < n; i += L)
            for (int j = i; j < i + (L >> 1); ++j)
                a[j + (L >> 1)] += a[j] * op;
}
const int N = 21;
int f[N+1][1<<N], g[N+1][1<<N], h[N+1][1<<N];
void subset_convolution(int *a,
    int *b, int *c, int L) {
    // c_k = \sum_{i+j=k, i&j=0} a_i * b_j
    int n = 1<<L;
    for (int i = 0; i <= L; ++i)
        fill(f[i], f[i] + n, 0), fill(g[i], g[i] + n, 0),
        fill(h[i], h[i] + n, 0);
    for (int i = 0; i < n; ++i)
        f[popcount((unsigned)i)][i] = a[i],
        g[popcount((unsigned)i)][i] = b[i];
    for (int i = 0; i <= L; ++i)
        fwt(f[i], n, 1), fwt(g[i], n, 1);
    for (int i = 0; i <= L; ++i)
        for (int j = 0; j <= i; ++j)
            for (int x = 0; x < n; ++x)
                h[i][x] += f[j][x] * g[i-j][x];
    for (int i = 0; i <= L; ++i)
        fwt(h[i], n, -1);
    for (int i = 0; i < n; ++i)
        c[i] = h[popcount((unsigned)i)][i];
}
Usage:
fwt(a, n, 1); fwt(b, n, 1); // n = 2^k
for (int i = 0; i < n; ++i) a[i] *= b[i];
fwt(a, n, -1); // inv: or/and -1, xor 1/2
subset_convolution(a, b, c, L); // O(2^L L^2)

```

7.4 Polynomial Operation [40adc5]

Mul/ Inv/ Sqrt/ DivMod/ Ln/ Exp/ Pow/ Eval/ Interpolate over mod 998244353. Mostly $O(n \log n)$, Eval $O(n \log^2 n)$.

```

#define
fi(s, n) for (int i = (int)(s); i < (int)(n); ++i)
template<int MAXN, ll P, ll RT> // MAXN = 2^k
struct Poly : vector<ll> { // coefficients in [0, P)
    using vector<ll>::vector;
    static NT<MAXN, P, RT> ntt;
    int n() const { return (int)size(); } // n() >= 1
    Poly(const Poly &p, int m) : vector<ll>(m) {
        copy_n(p.data(), min(p.n(), m), data());
    }
    Poly& irev() {
        return reverse(data(), data() + n()), *this;
    }
    Poly& isz(int m) { return resize(m), *this; }
    Poly& iadd(const Poly &rhs) { // n() == rhs.n()
        fi(0, n()) if (((*this)[i] += rhs[i]) >= P)
            (*this)[i] -= P;
        return *this;
    }
    Poly& imul(ll k) {
        fi(0, n()) (*this)[i] = (*this)[i] * k % P;
        return *this;
    }
    Poly Mul(const Poly &rhs) const {
        int m = 1;
        while (m < n() + rhs.n() - 1) m <=<= 1;
        Poly X(*this, m), Y(rhs, m);
        ntt(X.data(), m), ntt(Y.data(), m);
        fi(0, m) X[i] = X[i] * Y[i] % P;
        ntt(X.data(), m, true);
        return X.isz(n() + rhs.n() - 1);
    }
    Poly Inv() const { // (*this)[0] != 0, 1e5/95ms
        if (n() == 1) return {ntt.minv((*this)[0])};
        int m = 1;
        while (m < n() * 2) m <=<= 1;
        Poly Xi = Poly(*this, (n() + 1) / 2).Inv().isz(m);
        Poly Y(*this, m);
        ntt(Xi.data(), m), ntt(Y.data(), m);
        fi(0, m) {
            Xi[i] *= (2 - Xi[i] * Y[i]) % P;
            if ((Xi[i] % P) < 0) Xi[i] += P;
        } ntt(Xi.data(), m, true);
        return Xi.isz(n());
    }
    Poly Sqrt() const { // Jacobi(t[0], P) = 1, 1e5/235ms
        if (n() == 1) return {
            QuadraticResidue((*this)[0], P)};
    }
}

```



```

Poly X = Poly(*this,
  (n() + 1) / 2).Sqrt().isz(n());
return X.iadd(Mul(X.Inv()).isz(n())).imul(P /
  2 + 1);
}
pair<Poly, Poly> DivMod(const Poly &rhs) const {
  if (n() < rhs.n()) return {{0}, *this};
  // back() != 0
  const int m = n() - rhs.n() + 1;
  Poly X(rhs); X.irev().isz(m);
  Poly Y(*this); Y.irev().isz(m);
  Poly Q = Y.Mul(X.Inv()).isz(m).irev();
  X = rhs.Mul(Q), Y = *this;
  fi(0, n()) if ((Y[i] - X[i]) < 0) Y[i] += P;
  return {Q, Y.isz(max(1, rhs.n() - 1))};
}
Poly Dx() const {
  Poly ret(n() - 1);
  fi(0,
    ret.n()) ret[i] = (i + 1) * (*this)[i + 1] % P;
  return ret.isz(max(1, ret.n()));
}
Poly Sx() const {
  Poly ret(n() + 1);
  fi(0, n()) ret[i + 1] =
    ntt.minv(i + 1) * (*this)[i] % P;
  return ret;
}
Poly _tmul(int nn, const Poly &rhs) const {
  Poly Y = Mul(rhs).isz(n() + nn - 1);
  return Poly(Y.data() + n() - 1, Y.data() + Y.n());
}
vector<ll> _eval(const vector<ll> &x,
  const vector<Poly> &up) const {
  const int m = (int)x.size();
  if (!m) return {};
  vector<Poly> down(m * 2);
  down[1] =
    Poly(up[1].irev().isz(n()).Inv().irev()._tmul(m,
      *this);
  fi(2, m * 2)
    down[i] = up[i ^ 1]._tmul(up[i].n() - 1,
      down[i / 2]);
  vector<ll> y(m);
  fi(0, m) y[i] = down[m + i][0];
  return y;
}
static vector<Poly> _tree1(const vector<ll> &x) {
  const int m = (int)x.size();
  vector<Poly> up(m * 2);
  fi(0, m) up[m + i] = {(x[i] ? P - x[i] : 0), 1};
  for (int i = m - 1; i > 0; --i)
    up[i] = up[i * 2].Mul(up[i * 2 + 1]);
  return up;
}
vector<ll> Eval(const vector<ll> &x) const {
  // 1e5, 1s
  return _eval(x, _tree1(x));
}
static Poly Interpolate(const vector<ll> &x,
  const vector<ll> &y) { // 1e5, 1.4s
  const int m = (int)x.size();
  vector<Poly> up = _tree1(x), down(m * 2);
  vector<ll> z = up[1].Dx()._eval(x, up);
  fi(0, m) z[i] = y[i] * ntt.minv(z[i]) % P;
  fi(0, m) down[m + i] = {z[i]};
  for (int i = m - 1; i > 0; --i)
    down[i] = down[i * 2].Mul(up[i * 2 + 1])
      .iadd(down[i * 2 + 1].Mul(up[i * 2]));
  return down[1];
}
Poly Ln() const // (*this)[0] == 1, 1e5/170ms
{ return Dx().Mul(Inv()).Sx().isz(n()); }
Poly Exp() const { // (*this)[0] == 0, 1e5/360ms
  if (n() == 1) return {1};
  Poly X = Poly(*this, (n() + 1) / 2).Exp().isz(n());
  Poly Y = X.Ln(); Y[0] = P - 1;
  fi(0, n())
    if ((Y[i] - (*this)[i] - Y[i]) < 0) Y[i] += P;
  return X.Mul(Y).isz(n());
} // M := P(P - 1). If k >= M, k := k % M + M.
Poly Pow(ll k) const {
  int nz = 0;
  while (nz < n() && !(*this)[nz]) ++nz;
  if (nz * min(k, (ll)n()) >= n()) return Poly(n());
  if (!k) return Poly(Poly{1}, n());

```

```

Poly X(data() + nz, data() + nz + n() - nz * k);
const ll c = ntt.mpow(X[0], k % (P - 1));
return X.Ln().imul(k % P).Exp().imul(c)
  .irev().isz(n()).irev();
} // a_n = \sum c_j a_{n-j}
static ll LinearRecursion(const vector<ll> &a,
  const vector<ll> &coef, ll n) {
  const int k = (int)a.size();
  assert((int)coef.size() == k + 1);
  Poly C(k + 1), W(Poly{1}, k), M = {0, 1};
  fi(1, k + 1) C[k - i] = coef[i] ? P - coef[i] : 0;
  C[k] = 1;
  while (n) {
    if (n % 2) W = W.Mul(M).DivMod(C).second;
    n /= 2, M = M.Mul(M).DivMod(C).second;
  } ll ret = 0;
  fi(0, k) ret = (ret + W[i] * a[i]) % P;
  return ret;
}
};
#undef fi
using Poly_t = Poly<131072 * 2, 998244353, 3>;
template<> decltype(Poly_t::ntt) Poly_t::ntt = {};
Usage:
Poly_t f{1, 2, 3}, g{4, 5};
auto h = f.Mul(g), fi = f.Inv(); // f[0] != 0
auto q = f.Ln(); // f[0] == 1
auto e = g.Exp(); // g[0] == 0
auto y = f.Eval({1, 2, 3}); // multipoint

```

7.5 Value Polynomial [6788b6]

Polynomial held as consecutive point values; Lagrange eval and prefix-sum lift in $O(n)$.

```

struct Poly {
  mint base; // f(x) = poly[x - base]
  vector<mint> poly;
  Poly(mint b = 0, mint x = 0): base(b), poly(1, x) {}
  mint get_val(const mint &x) {
    if (x >= base && x < base + SZ(poly))
      return poly[(x - base).v];
    mint rt = 0;
    vector<mint> lmul(SZ(poly), 1), rmul(SZ(poly), 1);
    for (int i = 1; i < SZ(poly); ++i) lmul[i] =
      lmul[i - 1] * (x - (base + i - 1));
    for (int i = SZ(poly) - 2; i >= 0; --i)
      rmul[i] = rmul[i + 1] * (x - (base + i + 1));
    for (int i = 0; i < SZ(poly); ++i)
      rt += poly[i] * ifac[i] * inegfac[SZ(poly) -
        1 - i] * lmul[i] * rmul[i];
    return rt;
  }
  void raise() { // g(x) = sigma{base:x} f(x)
    if (SZ(poly) == 1 && poly[0] == 0)
      return;
    mint nw = get_val(base + SZ(poly)); poly.pb(nw);
    for (int i = 1; i < SZ(poly); ++i)
      poly[i] += poly[i - 1];
  }
};
Usage:
Poly f(base, y0); // f(base+i) = poly[i]
f.poly = {y0, y1, y2}; // consecutive values
mint v = f.get_val(x); // any x, O(n)
f.raise(); // f becomes prefix sum, deg+1

```

7.6 Newton's Method

Given $F(x)$ where

$$F(x) = \sum_{i=0}^{\infty} \alpha_i (x - \beta)^i$$

for β being some constant. Polynomial P such that $F(P) = 0$ can be found iteratively. Denote by Q_k the polynomials such that $F(Q_k) = 0 \pmod{x^{2^k}}$, then

$$Q_{k+1} = Q_k - \frac{F(Q_k)}{F'(Q_k)} \pmod{x^{2^{k+1}}}$$

8 Geometry

8.1 Basic [be1f06]

Point/line primitives the rest of this section depends on: * is dot, ^ is cross; project, reflect, segment inside, SegmentIntersect (endpoints included). Intersect divides by zero on parallel lines.

```

bool same(double a, double b) {
  return abs(a - b) < eps;
}
struct P {
  double x, y;
  P(): x(0), y(0) {}
  P(double x, double y): x(x), y(y) {}
  P operator + (P b) const { return {x + b.x, y + b.y};

```



```

    }
    P operator - (P b) const { return {x - b.x, y - b.y}; }
    }
    P operator * (double b) const {
        return {x * b, y * b}; }
    P operator / (double b) const {
        return {x / b, y / b}; }
    double operator * (P b) const {
        return x * b.x + y * b.y; }
    double operator ^ (P b) const {
        return x * b.y - y * b.x; }
    double abs() const { return hypot(x, y); }
    P unit() const { return *this / abs(); }
    P rot(double o) const {
        double c = cos(o), s = sin(o);
        return {c * x - s * y, s * x + c * y}; }
    }
    double angle() const { return atan2(y, x); }
};

struct L { // ax + by + c = 0
    double a, b, c, o; P pa, pb;
    L() : a(0), b(0), c(0), o(0), pa(), pb() {}
    L(P pa, P pb) : a(pa.y - pb.y),
        b(pb.x - pa.x), c(pa ^ pb),
        o(atan2(-a, b)), pa(pa), pb(pb) {}
    P project(P p) { P d = pb - pa;
        return pa + d * (d * (p - pa) / (d * d)); }
    P reflect(P p) { return p + (project(p) - p) * 2; }
    double get_ratio(P p) { P d = pb - pa;
        return (p - pa) * d / (d * d); }
    bool inside(P p) { return
        min(pa.x, pb.x) <= p.x && p.x <= max(pa.x, pb.x) &&
        min(pa.y, pb.y) <= p.y && p.y <= max(pa.y, pb.y) &&
        same(a * p.x + b * p.y, -c); }
};

bool SegmentIntersect(P p1, P p2, P p3, P p4) {
    if (max(p1.x, p2.x) < min(p3.x, p4.x) ||
        max(p3.x, p4.x) < min(p1.x, p2.x)) return false;
    if (max(p1.y, p2.y) < min(p3.y, p4.y) ||
        max(p3.y, p4.y) < min(p1.y, p2.y)) return false;
    return sign((p3 - p1) ^ (p4 - p1)) *
        sign((p3 - p2) ^ (p4 - p2)) <= 0 &&
        sign((p1 - p3) ^ (p2 - p3)) *
        sign((p1 - p4) ^ (p2 - p4)) <= 0;
}

bool parallel(L x, L y) {
    return same(x.a * y.b, x.b * y.a); }
P Intersect(L x, L y) {
    return P(-x.b * y.c + x.c * y.b,
        x.a * y.c - x.c * y.a)
        / (-x.a * y.b + x.b * y.a); }

```

8.2 Sector Area [6fba33]

Area of the sector between two vectors, $O(1)$. Returns the minor arc — subtract from the full disc for the major one.

```

// calc area of sector which include a, b
double SectorArea(P a, P b, double r) {
    double o = abs(remainder(a.angle() - b.angle(),
        2 * pi));
    return r * r * o / 2;
}

```

8.3 Half Plane Intersection [b25437]

Intersection polygon of half planes in $O(n \log n)$; the feasible side is to the left of each line. Empty result means degenerate or infeasible.

```

bool jizz(L l1, L l2, L l3) { P p = Intersect(l2, l3);
    return ((l1.pb - l1.pa) ^ (p - l1.pa)) < -eps; }

bool cmp(const L &a, const L &b) { return same(a.o,
    b.o) ? (((b.pb - b.pa) ^ (a.pb - b.pa)) > eps) : a.o < b.o; }

// available area for L l is (l.pb - l.pa) ^ (p - l.pa) > 0
vector<P> HPI(vector<L> &ls) {
    sort(ls.begin(), ls.end(), cmp);
    vector<L> pls(1, ls[0]);
    for(int i=0; i<(int)ls.size(); ++i) if(!same(ls[i].o,
        pls.back().o)) pls.push_back(ls[i]);
    deque<int> dq = {0, 1};
#define meow(a,b,c
    ) while(dq.size() > 1u && jizz(pls[a], pls[b], pls[c]))
    for(int i=2; i<(int)pls.size(); ++i) {
        meow(i, dq.back(), dq[dq.size() - 2]) dq.pop_back();
        meow(i, dq[0], dq[1]) dq.pop_front();
        dq.push_back(i);
    }
    meow(dq.front(), dq.back(),
        dq[dq.size() - 2]) dq.pop_back();
}

```

```

meow(dq.back(), dq[0], dq[1]) dq.pop_front();
if(dq.size() < 3u) return {};
// no solution or solution is not a convex
vector<P> rt; for(int i=0; i<(int)dq.size(); ++i)
    rt.push_back(Intersect(pls[dq[i]],
        pls[dq[(i+1)%dq.size()]]));
return rt;
}

```

Usage:

```

vector<L> hs = {L({0,0},{1,0}), L({1,0},{1,1}),
    L({1,1},{0,1}), L({0,1},{0,0})};
auto poly = HPI(hs); // unit square; left side is feasible

```

8.4 Rotating Sweep Line [39bc8f]

Sweeps all $\binom{n}{2}$ directions in $O(n^2 \log n)$; rotatingSweepLineOrder returns the final order, while the legacy void wrapper preserves the old call shape.

```

vector<int> rotatingSweepLineOrder(const
    vector<pair<int, int>> &ps) {
    int n = (int)ps.size(); if (!n) return {};
    vector<int> id(n), pos(n);
    vector<pair<int, int>> line(n*(n-1)/2);
    int m = 0;
    for(int i=0; i<n; ++i) for(int j=i+1; j<n; ++j) line[m++] = {
        i, j};
    sort(line.begin(), line.end(), [&](const pair<int,
        int> &a, const pair<int, int> &b) {
        long long ax = ps[a.first].first - ps[a.second].first;
        long long ay = ps[a.first].second -
            ps[a.second].second;
        long long bx = ps[b.first].first - ps[b.second].first;
        long long by = ps[b.first].second -
            ps[b.second].second;
        if (ax == 0 || bx == 0) return ax == 0 && bx != 0;
        return ay * bx < by * ax;
    });
    iota(id.begin(), id.end(), 0);
    sort(id.begin(), id.end(), [&](const int &a,
        const int &b) { return ps[a] < ps[b]; });
    for(int i=0; i<n; ++i) pos[id[i]] = i;

    for(int i=0; i<m; ++i) {
        auto l = line[i];
        tie(pos[l.first], pos[l.second], id[pos[l.first]],
            id[pos[l.second]]) = make_tuple(pos[l.second],
                pos[l.first], l.second, l.first);
    }
    return id;
}

void rotatingSweepLine(vector<pair<int, int>> &ps) {
    (void) rotatingSweepLineOrder(ps);
}

```

Usage:

```

vector<pair<int, int>> p = {{0,0},{2,0},{1,1}};
vector<int> order = rotatingSweepLineOrder(p); // point ids

```

8.5 Triangle Center [fdd4fc]

Circumcentre, centroid, orthocentre (via the Euler line) and incentre, each $O(1)$. Degenerate (collinear) input divides by zero.

```

Point TriangleCircumCenter(Point a, Point b, Point c) {
    Point u = b - a, v = c - a;
    double x = u.x * u.x + u.y * u.y,
        y = v.x * v.x + v.y * v.y;
    double d = 2 * Cross(u, v);
    return a + Point((v.y * x - u.y * y) / d,
        (u.x * y - v.x * x) / d);
}

Point TriangleMassCenter(Point a, Point b, Point c) {
    return (a + b + c) / 3.0;
}

Point TriangleOrthoCenter(Point a, Point b, Point c) {
    return TriangleMassCenter(a, b,
        c) * 3.0 - TriangleCircumCenter(a, b, c) * 2.0;
}

Point TriangleInnerCenter(Point a, Point b, Point c) {
    double la = len(b - c);
    double lb = len(a - c);
    double lc = len(a - b);
    double s = la + lb + lc;
    return {(la * a.x + lb * b.x + lc * c.x) / s,
        (la * a.y + lb * b.y + lc * c.y) / s};
}

```

8.6 Polygon Center [7477e0]

Area centroid of a simple polygon in $O(n)$ by signed-area weighting, so concave shapes are fine.

```
Point BaryCenter(vector<Point> &p, int n) {
    Point res(0, 0);
    double s = 0, t;
    for (int i = 1; i < p.size() - 1; i++) {
        t = Cross(p[i] - p[0], p[i + 1] - p[0]) / 2;
        s += t;
        res.x += (p[0].x + p[i].x + p[i + 1].x) * t;
        res.y += (p[0].y + p[i].y + p[i + 1].y) * t;
    } res.x /= 3 * s, res.y /= 3 * s;
    return res;
}
```

8.7 Maximum Triangle [dc6c45]

Largest triangle inscribed in a convex polygon by exhaustive triples in $O(n^3)$.

```
double ConvexHullMaxTriangleArea(Point p[],
    int res[], int chnum) {
    double area = 0;
    for (int i = 0; i < chnum; ++i)
        for (int j = i + 1; j < chnum; ++j)
            for (int k = j + 1; k < chnum; ++k)
                area = max(area, fabs(Cross(p[res[j]] - p[res[i]], p[res[k]] - p[res[i]])));
    return area / 2;
}
```

8.8 Point in Polygon [0a9a66]

Ray casting for any simple polygon, $O(n)$. Returns 1 on the boundary, 2 strictly inside, 0 outside.

```
int pip(vector<P> ps, P p) {
    int c = 0;
    for (int i = 0; i < ps.size(); ++i) {
        int a = i, b = (i + 1) % ps.size();
        L l(ps[a], ps[b]);
        P q = l.project(p);
        if ((p - q).abs() < eps && l.inside(q)) return 1;
        if (same(ps[a].y, ps[b].y) &&
            same(ps[a].y, p.y)) continue;
        if (ps[a].y > ps[b].y) swap(a, b);
        if (ps[a].y <= p.y && p.y < ps[b].y &&
            p.x <= ps[a].x + (ps[b].x - ps[a].x) /
            (ps[b].y - ps[a].y) * (p.y - ps[a].y)) ++c;
    } return (c & 1) * 2;
}
```

Usage:
vector<P> poly = {{0,0},{4,0},{4,4},{0,4}};
int where = pip(poly, q); // 0 out, 1 boundary, 2 in

8.9 Circle [91e2d1]

Circle intersection points and area, circle-segment crossing, and circle $\cap$ triangle area. All $O(1)$.

```
struct C {
    P c; double r;
    C(P c = P(0, 0), double r = 0) : c(c), r(r) {}
};

vector<P> Intersect(C a, C b) {
    if (a.r > b.r) swap(a, b);
    double d = (a.c - b.c).abs();
    vector<P> p;
    if (same(a.r + b.r, d))
        p.push_back(a.c + (b.c - a.c).unit() * a.r);
    else if (a.r + b.r > d && d + a.r >= b.r) {
        double o = acos((sq(a.r) + sq(d) - sq(b.r)) / (2 * a.r * d));
        P i = (b.c - a.c).unit();
        p.push_back(a.c + i.rot(o) * a.r);
        p.push_back(a.c + i.rot(-o) * a.r);
    } return p;
}

double IntersectArea(C a, C b) {
    if (a.r > b.r) swap(a, b);
    double d = (a.c - b.c).abs();
    if (d >= a.r + b.r - eps) return 0;
    if (d + a.r <= b.r + eps) return sq(a.r) * acos(-1);
    double p = acos((sq(a.r) + sq(d) - sq(b.r)) / (2 * a.r * d));
    double q = acos((sq(b.r) + sq(d) - sq(a.r)) / (2 * b.r * d));
    return p * sq(a.r) + q * sq(b.r) - a.r * d * sin(p);
}

// drop 2nd level to get points for a line (default: segment)
vector<P> CircleCrossLine(P a, P b, P o, double r) {
    double x = b.x - a.x,
        y = b.y - a.y, A = sq(x) + sq(y);
    double B = 2 * x * (a.x - o.x) + 2 * y * (a.y - o.y);
```

```
double C = sq(a.x - o.x) + sq(a.y - o.y) - sq(r);
double d = B * B - 4 * A * C;
vector<P> t;
if (d >= -eps) {
    d = max(0., d);
    double i = (-B - sqrt(d)) / (2 * A);
    double j = (-B + sqrt(d)) / (2 * A);
    if (i - 1.0 <= eps && i >= -eps)
        t.emplace_back(a.x + i * x, a.y + i * y);
    if (j - 1.0 <= eps && j >= -eps)
        t.emplace_back(a.x + j * x, a.y + j * y);
} return t;
}

// area of circle(r) cap triangle OAB
double AreaOfCircleTriangle(P a, P b, double r) {
    bool ina = a.abs() < r, inb = b.abs() < r;
    auto p = CircleCrossLine(a, b, P(0, 0), r);
    if (ina) {
        if (inb) return abs(a ^ b) / 2;
        return SectorArea(b, p[0], r) + abs(a ^ p[0]) / 2;
    }
    if (inb) return SectorArea(p[0], a, r) + abs(p[0] ^ b) / 2;
    if (p.size() == 2u)
        return SectorArea(a, p[0], r) +
            SectorArea(p[1], b, r) + abs(p[0] ^ p[1]) / 2;
    return SectorArea(a, b, r);
}
```

// for any triangle

```
double AreaOfCircleTriangle(vector<P> ps, double r) {
    double ans = 0;
    for (int i = 0; i < 3; ++i) {
        int j = (i + 1) % 3;
        double o = atan2(ps[i].y, ps[i].x) - atan2(ps[j].y, ps[j].x);
        if (o >= pi) o = o - 2 * pi;
        if (o <= -pi) o = o + 2 * pi;
        ans += AreaOfCircleTriangle(ps[i], ps[j], r) * (o >= 0 ? 1 : -1);
    } return abs(ans);
}
```

8.10 Tangent of Circles and Points to Circle [c3609c]

Common tangents of two circles (0/1/2/3/4 by configuration) and tangents from a point, $O(1)$.

```
vector<L> tangent(C a, C b) {
#define Pij \
    P i = (b.c - a.c).unit() * a.r, j = P(i.y, -i.x);\
    z.emplace_back(a.c + i, a.c + i + j);
#define deo(I,J) \
    double e = a.r I b.r, o = acos(e / d);\
    P i = \
        (b.c - a.c).unit(), j = i.rot(o), k = i.rot(-o);\
    z.emplace_back(a.c + j * a.r, b.c J j * b.r);\
    z.emplace_back(a.c + k * a.r, b.c J k * b.r);
    if (a.r < b.r) swap(a, b);
    vector<L> z;
    double d = (a.c - b.c).abs();
    if (d + b.r < a.r) return z;
    else if (same(d + b.r, a.r)) { Pij; }
    else {
        deo(+,+);
        if (same(d, a.r + b.r)) { Pij; }
        else if (d > a.r + b.r) { deo(+,-); }
    } return z;
}
```

```
vector<L> tangent(C c, P p) {
    vector<L> z;
    double d = (p - c.c).abs();
    if (same(d, c.r)) {
        P i = (p - c.c).rot(pi / 2);
        z.emplace_back(p, p + i);
    } else if (d > c.r) {
        double o = acos(c.r / d);
        P i = (p - c.c).unit(),
            j = i.rot(o) * c.r, k = i.rot(-o) * c.r;
        z.emplace_back(c.c + j, p);
        z.emplace_back(c.c + k, p);
    } return z;
}
```

8.11 Area of Union of Circles [490636]

Union area by integrating uncovered arcs with Green's theorem, $O(n^2 \log n)$. Duplicate circles must be removed first or the area is double counted.

```
vector<pair<double, double>> CoverSegment(C &a, C &b) {
```

```

double d = (a.c - b.c).abs();
vector<pair<double, double>> res;
if (same(a.r + b.r, d)) ;
else if (d <= abs(a.r - b.r) + eps) {
    if (a.r < b.r) res.emplace_back(0, 2 * pi);
} else if (d < abs(a.r + b.r) - eps) {
    double o = acos((sq(a.r) + sq(d) -
        sq(b.r)) / (2 * a.r * d)),
        z = (b.c - a.c).angle();
    if (z < 0) z += 2 * pi;
    double l = z - o, r = z + o;
    if (l < 0) l += 2 * pi;
    if (r > 2 * pi) r -= 2 * pi;
    if (l > r) res.emplace_back(l,
        2 * pi), res.emplace_back(0, r);
    else res.emplace_back(l, r);
} return res;
}

double CircleUnionArea(vector<C> c) {
// circle should be identical
int n = c.size();
double a = 0, w;
for (int i = 0; w = 0, i < n; ++i) {
    vector<pair<double, double>> s = {{2 * pi, 9}}, z;
    for (int j = 0; j < n; ++j) if (i != j) {
        z = CoverSegment(c[i], c[j]);
        for (auto &e : z) s.push_back(e);
    } sort(s.begin(), s.end());
    auto F = [&](double t) {
        return c[i].r * (c[i].r * t
            + c[i].c.x * sin(t) - c[i].c.y * cos(t));
    };
    for (auto &e : s) {
        if (e.first > w) a += F(e.first) - F(w);
        w = max(w, e.second);
    }
} return a * 0.5;
}

```

8.12 Minimum Distance of 2 Polygons [dbe093]

Minimum distance of two convex polygons. The self-contained implementation checks edge pairs and containment in $O(nm)$, avoiding hidden distance-helper contracts.

```

// Convex polygons in cyclic order; self-contained
// O(nm) implementation.
double _point_seg_dist(Point p, Point a, Point b) {
    double dx = b.x - a.x, dy = b.y - a.y,
        z = dx * dx + dy * dy;
    double t = z ? clamp(((p.x - a.x) * dx +
        (p.y - a.y) * dy) / z, 0., 1.) : 0;
    return hypot(p.x - (a.x + t * dx),
        p.y - (a.y + t * dy));
}

double _cross(Point a, Point b, Point c) {
    return (b.x - a.x) * (c.y - a.y) -
        (b.y - a.y) * (c.x - a.x);
}

bool _on_segment(Point a, Point b, Point p) {
    return fabs(_cross(a, b, p)) <= 1e-10 &&
        min(a.x, b.x) - 1e-10 <= p.x &&
        p.x <= max(a.x, b.x) + 1e-10 &&
        min(a.y, b.y) - 1e-10 <= p.y &&
        p.y <= max(a.y, b.y) + 1e-10;
}

bool _segments_intersect(Point a,
    Point b, Point c, Point d) {
    double x1 = _cross(a, b, c), x2 = _cross(a, b, d);
    double x3 = _cross(c, d, a), x4 = _cross(c, d, b);
    if (_on_segment(a, b, c) || _on_segment(a, b, d) ||
        _on_segment(c, d, a) ||
        _on_segment(c, d, b)) return true;
    return (x1 > 0) != (x2 > 0) && (x3 > 0) != (x4 > 0);
}

bool _inside_convex(Point p, Point poly[], int n) {
    int sign = 0;
    for (int i = 0; i < n; ++i) {
        double c = _cross(poly[i], poly[(i + 1) % n], p);
        if (fabs(c) <= 1e-10) continue;
        if (!sign) sign = c > 0 ? 1 : -1;
        else if ((c > 0 ? 1 : -1) != sign) return false;
    }
    return true;
}

```

```

double TwoConvexHullMinDist(Point P[],
    Point Q[], int n, int m) {
    double ans = 1e100;
    for (int i = 0; i < n; ++i) {
        Point a = P[i], b = P[(i + 1) % n];
        for (int j = 0; j < m; ++j) {
            Point c = Q[j], d = Q[(j + 1) % m];
            if (_segments_intersect(a, b, c, d)) return 0;
            ans = min(ans, _point_seg_dist(a, c, d));
            ans = min(ans, _point_seg_dist(c, a, b));
        }
    }
    if (_inside_convex(P[0], Q, m) ||
        _inside_convex(Q[0], P, n)) return 0;
    return ans;
}

```

8.13 2D Convex Hull [cedfb3]

Monotone chain in $O(n \log n)$ (collinear points dropped), plus CH for $O(\log n)$ extreme-point, line-intersection, containment and tangent queries.

```

bool operator<(const P &a, const P &b) {
    return same(a.x, b.x) ? a.y < b.y : a.x < b.x;
}
bool operator>(const P &a, const P &b) {
    return same(a.x, b.x) ? a.y > b.y : a.x > b.x;
}
#define crx(a, b, c) ((b - a) ^ (c - a))
vector<P> convex(vector<P> ps) {
    if (ps.empty()) return {};
    vector<P> p;
    sort(ps.begin(), ps.end());
    for (int i = 0; i < ps.size(); ++i) {
        while (p.size() >= 2 &&
            crx(p[p.size() - 2], ps[i],
                p[p.size() - 1]) >= 0)
            p.pop_back();
        p.push_back(ps[i]);
    }
    int t = p.size();
    for (int i = (int)ps.size() - 2; i >= 0; --i) {
        while (p.size() > t &&
            crx(p[p.size() - 2], ps[i],
                p[p.size() - 1]) >= 0)
            p.pop_back();
        p.push_back(ps[i]);
    }
    p.pop_back();
    return p;
}

int sgn(double x) {
    return same(x, 0) ? 0 : x > 0 ? 1 : -1;
}
P isLL(P p1, P p2, P q1, P q2) {
    double a = crx(q1, q2, p1), b = -crx(q1, q2, p2);
    return (p1 * b + p2 * a) / (a + b);
}

struct CH {
    int n;
    vector<P> p, u, d;
    CH() {}
    CH(vector<P> ps) : n(ps.size()), p(ps) {
        rotate(p.begin(), min_element(p.begin(),
            p.end()), p.end());
        auto t = max_element(p.begin(), p.end());
        d = vector<P>(p.begin(), next(t));
        u = vector<P>(t, p.end()); u.push_back(p[0]);
    }
    int find(vector<P> &v, P d) {
        int l = 0, r = v.size();
        while (l + 5 < r) {
            int L = (l * 2 + r) / 3, R = (l + r * 2) / 3;
            if (v[L] * d > v[R] * d) r = R;
            else l = L;
        }
        int x = l;
        for (int i = l + 1; i < r; ++i)
            if (v[i] * d > v[x] * d) x = i;
        return x;
    }
    int findFarest(P v) {
        if (v.y > 0 || v.y == 0 && v.x > 0)
            return ((int)d.size() - 1 + find(u,
                v)) % p.size();
        return find(d, v);
    }
}

P get(int l, int r, P a, P b) {
    int s = sgn(crx(a, b, p[l % n]));
    while (l + 1 < r) {
        int m = (l + r) >> 1;
        if (sgn(crx(a, b, p[m % n])) == s) l = m;
        else r = m;
    }
    return isLL(a, b, p[l % n], p[(l + 1) % n]);
}

```

```

}
vector<P> getLineIntersect(P a, P b) {
    int X = findFarest((b - a).rot(pi / 2));
    int Y = findFarest((a - b).rot(pi / 2));
    if (X > Y) swap(X, Y);
    if (sgn(crx(a, b,
        p[X])) * sgn(crx(a, b, p[Y])) < 0)
        return {get(X, Y, a, b), get(Y, X + n, a, b)};
    return {}; // tangent case falls here
}
void update_tangent(P q, int i, int &a, int &b) {
    if (sgn(crx(q, p[a], p[i])) > 0) a = i;
    if (sgn(crx(q, p[b], p[i])) < 0) b = i;
}
void bs(int l, int r, P q, int &a, int &b) {
    if (l == r) return;
    update_tangent(q, l % n, a, b);
    int s = sgn(crx(q, p[l % n], p[(l + 1) % n]));
    while (l + 1 < r) {
        int m = (l + r) >> 1;
        if (sgn(crx(q, p[m % n],
            p[(m + 1) % n])) == s) l = m;
        else r = m;
    } update_tangent(q, r % n, a, b);
}
bool contain(P p) {
    if (p.x < d[0].x || p.x > d.back().x) return 0;
    auto it = lower_bound(d.begin(),
        d.end(), P(p.x, -1e12));
    if (it->x == p.x) {
        if (it->y > p.y) return 0;
    } else if (crx(*prev(it), *it, p) < -eps) return 0;
    it = lower_bound(u.begin(), u.end(),
        P(p.x, 1e12), greater<P>());
    if (it->x == p.x) {
        if (it->y < p.y) return 0;
    } else if (crx(*prev(it), *it, p) < -eps) return 0;
    return 1;
}
bool get_tangent(P p, int &a, int &b) { // b -> a
    if (!contain(p)) return 0;
    a = b = 0;
    int i = lower_bound(d.begin(),
        d.end(), p) - d.begin();
    bs(0, i, p, a, b);
    bs(i, d.size(), p, a, b);
    i = lower_bound(u.begin(), u.end(),
        p, greater<P>()) - u.begin();
    bs((int)d.size() - 1,
        (int)d.size() - 1 + i, p, a, b);
    bs((int)d.size() - 1 + i,
        (int)d.size() - 1 + u.size(), p, a, b);
    return 1;
}
};
Usage:
vector<P> hull = convex(points); // CCW; drops collinear points
CH ch(hull); bool inside = ch.contain(q);

```

8.14 3D Convex Hull [29e4a9]

Incremental 3D hull in $O(n^2)$; faces() merges coplanar triangles. **Input must not contain duplicate points.**

```

double absvol(const P a, const P b, const P c, const P d)
{ return abs(((b-a)^(c-a))*(d-a))/6; }
struct convex3D {
    static const int maxn=1010;
    struct T{
        int a,b,c; bool res; T(){
            T(int a,int b,int c,bool res=1):a(a),
                b(b),c(c),res(res){}
        };
    int n,m;
    P p[maxn];
    T f[maxn*8];
    int id[maxn][maxn];
    bool on(T &t, P &q) { return ((p[t.c]-
        p[t.b])^(p[t.a]-p[t.b]))*(q-p[t.a])>eps; }
    void meow(int q,int a,int b){
        int g=id[a][b];
        if (f[g].res){
            if (on(f[g], p[q])) dfs(q,g);
            else{
                id[q][b]=id[a][q]=id[b][a]=m;
                f[m++]=T(b,a,q,1);
            }
        }
    }
};

```

```

void dfs(int p,int i){ f[i].res=0;
    meow(p,f[i].b,f[i].a);
    meow(p,f[i].c,f[i].b); meow(p,f[i].a,f[i].c); }
void operator()(){
    if (n<4) return;
    if ([&](){
        for (int i=1;i<n;++i) if (abs(p[0]-p[i])>eps)
            return swap(p[1],p[i]),0;
        return 1;
    }() || [&](){
        for (int i=2;i<n;++i)
            if (abs((p[0]-p[i])^(p[1]-p[i]))>eps)
                return swap(p[2],p[i]),0;
        return 1;
    }() || [&](){
        for (int i=3;i<n;++i)
            if (abs(((p[1]-p[0])^(p[2]-
                p[0]))*(p[i]-p[0]))>eps)
                return swap(p[3],p[i]),0;
        return 1;
    }()) return;
    for (int i=0;i<4;++i){
        T t((i+1)%4,(i+2)%4,(i+3)%4,1);
        if (on(t,p[i])) swap(t.b,t.c);
        id[t.a][t.b]=id[t.b][t.c]=id[t.c][t.a]=m;
        f[m++]=t;
    } for (int i=4;i<n;++i) for (int j=0;j<m;++j)
        if (f[j].res && on(f[j],p[i])){ dfs(i,j); break; }
    int mm=m; m=0;
    for (int i=0;i<mm;++i) if (f[i].res) f[m++]=f[i];
}
bool same(int i,int j){
    return !(absvol(p[f[i].a],
        p[f[i].b],p[f[i].c],p[f[j].a])>eps
        || absvol(p[f[i].a],p[f[i].b],
            p[f[i].c],p[f[j].b])>eps
        || absvol(p[f[i].a],p[f[i].c],
            p[f[i].b],p[f[j].c])>eps);
}
int faces(){
    int r=0;
    for (int i=0;i<m;++i){
        int iden=1;
        for (int j=0;j<i;++j) if (same(i,j)) iden=0;
        r+=iden;
    } return r;
}
} tb;

```

8.15 Minimum Enclosing Circle [0d85a1]

Welzl randomized incremental with shuffle, expected $O(n)$.

```

pt center(const pt &a, const pt &b, const pt &c) {
    pt p0 = b - a, p1 = c - a;
    double c1 = norm2(p0) * 0.5, c2 = norm2(p1) * 0.5;
    double d = p0 ^ p1;
    double x = a.x + (c1 * p1.y - c2 * p0.y) / d;
    double y = a.y + (c2 * p0.x - c1 * p1.x) / d;
    return pt(x, y);
}

```

```

circle min_enclosing(vector<pt> &p) {
    static mt19937 rng(712367821);
    shuffle(p.begin(), p.end(), rng);
    double r = 0.0;
    pt cent;
    for (int i = 0; i < p.size(); ++i) {
        if (norm2(cent - p[i]) <= r) continue;
        cent = p[i];
        r = 0.0;
        for (int j = 0; j < i; ++j) {
            if (norm2(cent - p[j]) <= r) continue;
            cent = (p[i] + p[j]) / 2;
            r = norm2(p[j] - cent);
            for (int k = 0; k < j; ++k) {
                if (norm2(cent - p[k]) <= r) continue;
                cent = center(p[i], p[j], p[k]);
                r = norm2(p[k] - cent);
            }
        }
    } return circle(cent, sqrt(r));
}

```

Usage:
vector<pt> points = {{0,0},{2,0},{0,2}};
circle c = min_enclosing(points); // points are shuffled

8.16 Closest Pair [f6de57]

Divide and conquer in $O(n \log^2 n)$ (re-sorts the strip by y). $p[]$ must already be sorted by x .


```
double closest_pair(int l, int r) {
    // p should be sorted increasingly according to the
    // x-coordinates.
    if (l == r) return 1e9;
    if (r - l == 1) return dist(p[l], p[r]);
    int m = (l + r) >> 1;
    double d = min(closest_pair(l,
        m), closest_pair(m + 1, r));
    vector<int> vec;
    for (int i = m; i >= l &&
        fabs(p[m].x - p[i].x) < d; --i) vec.push_back(i);
    for (int i = m + 1; i <= r &&
        fabs(p[m].x - p[i].x) < d; ++i) vec.push_back(i);
    sort(vec.begin(), vec.end(), [&](int a, int b) {
        return p[a].y < p[b].y; });
    for (int i = 0; i < vec.size(); ++i) {
        for (int j = i + 1; j < vec.size() &&
            fabs(p[vec[j]].y - p[vec[i]].y) < d; ++j) {
            d = min(d, dist(p[vec[i]], p[vec[j]]));
        }
    }
    return d;
}

Usage:
sort(p, p+n, [](auto a, auto b) { return a.x < b.x; });
double d = closest_pair(0, n-1);
```

9 Else

9.1 Cyclic Ternary Search* [9017cc]

Minimum of a **cyclically shifted** unimodal array in $O(\log n)$; $\text{pred}(a, b)$ compares two positions.

```
/* bool pred(int a, int b);
f(0) ~ f(n - 1) is a cyclic-shift U-function
return idx s.t. pred(x, idx) is false forall x*/
int cyc_tsearch(int n, auto pred) {
    if (n == 1) return 0;
    int l = 0, r = n; bool rv = pred(1, 0);
    while (r - l > 1) {
        int m = (l + r) / 2;
        if (pred(0, m) ? rv : pred(m, (m + 1) % n)) r = m;
        else l = m;
    }
    return pred(1, r % n) ? l : r % n;
}

Usage:
vector<int> a = {3, 4, 5, 1, 2};
int at = cyc_tsearch(a.size(),
    [&](int i, int j) { return a[i] < a[j]; }); // at=3
```

9.2 Mo's Algorithm (With modification) [f47cfd]

Mo's with point updates, block $N^{2/3}$, $O(N^{5/3})$. The callback-based solve applies/rolls back updates and reports each query.

```
/*
Mo's algorithm with point updates.
'add_time(t, L, R)' applies update t,
'sub_time' rolls it back, and the other
callbacks maintain the current [L,
R] answer. Updates are numbered from 1.
*/
struct Query {
    int L, R, LBid, RBid, T, id;
    Query(int l, int r, int t,
        int block = 1, int query_id = -1)
        : L(l), R(r), LBid(l / block),
        RBid(r / block), T(t), id(query_id) {}
    bool operator<(const Query &q) const {
        return tie(LBid, RBid,
            T) < tie(q.LBid, q.RBid, q.T);
    }
};

template<class AddTime, class SubTime,
    class Add, class Sub, class Answer>
void solve(vector<Query> query,
    AddTime add_time, SubTime sub_time,
    Add add, Sub sub, Answer answer) {
    sort(query.begin(), query.end());
    int L = 0, R = -1, T = 0;
    for (const Query &q : query) {
        while (T < q.T) add_time(++T, L, R);
        while (T > q.T) sub_time(T--, L, R);
        while (R < q.R) add(++R);
        while (L > q.L) add(--L);
        while (R > q.R) sub(R-);
        while (L < q.L) sub(L++);
        answer(q);
    }
}

Usage:
```

```
int B = max(1, (int)pow(n, 2.0/3));
qs.emplace_back(l, r, updates_seen, B, id);
solve(qs,
    [&](int t, int L, int R) { apply_update(t, L, R); },
    [&](int t, int L, int R) { rollback_update(t, L, R); },
    [&](int i) { add_pos(i); }, [&](int i) { remove_pos(i); },
    [&](const Query &q) { ans[q.id] = current_answer(); });
```

9.3 Mo's Algorithm On Tree [0c6c06]

Path queries via the Euler bracket sequence, $O(N\sqrt{N})$; the callback-based constructor takes the caller's LCA and solve handles the extra LCA vertex.

```
/*
Mo's algorithm on tree paths
. 'in'/'out' are the entry/exit positions in a
2n Euler tour and 'ord' maps Euler
positions back to vertices.
*/
struct Query {
    int L, R, LBid, lca, id;
    template<class LCA>
    Query(int u, int v, int block, const vector<int> &in,
        const vector<int> &out, LCA get_lca,
        int query_id = -1) : id(query_id) {
        if (in[u] > in[v]) swap(u, v);
        int c = get_lca(u, v);
        if (c == u) L = in[u], R = in[v], lca = -1;
        else L = out[u], R = in[v], lca = c;
        LBid = L / block;
    }
    bool operator<(const Query &q) const {
        return tie(LBid, R) < tie(q.LBid, q.R);
    }
};

template<class Flip, class Answer>
void solve(vector<Query> query,
    const vector<int> &ord, Flip flip,
    Answer answer) {
    sort(query.begin(), query.end());
    int L = 0, R = -1;
    for (const Query &q : query) {
        while (R < q.R) flip(ord[++R]);
        while (L > q.L) flip(ord[--L]);
        while (R > q.R) flip(ord[R-]);
        while (L < q.L) flip(ord[L++]);
        if (q.lca != -1) flip(q.lca);
        answer(q);
        if (q.lca != -1) flip(q.lca);
    }
}

Usage:
int B = max(1, (int)sqrt(2*n));
qs.emplace_back(u, v, B, in, out, get_lca, id);
solve(qs, ord, [&](int v) { toggle(v); },
    [&](const Query &q) { ans[q.id] = current_answer(); });
```

9.4 Additional Mo's Algorithm Trick

• Mo's Algorithm With Addition Only

- Sort query same as the normal Mo's algorithm.
- For each query $[l, r]$:
 - If $l/blk = r/blk$, brute-force.
 - If $l/blk \neq curL/blk$, initialize $curL := (l/blk + 1) \cdot blk$, $curR := curL - 1$
 - If $r > curR$, increase $curR$
 - decrease $curL$ to fit l , and then undo after answering

• Mo's Algorithm With Offline Second Time

- Require: Changing answer $\equiv$ adding $f([l, r], r+1)$.
- Require: $f([l, r], r+1) = f([l, r], r) + f([l, l], r+1)$.
- Part 1: Answer all $f([l, r], r+1)$ first.
- Part 2: Store $curR \rightarrow R$ for $curL$ (reduce the space to $O(N)$), and then answer them by the second offline algorithm.
- Note: You must do the above symmetrically for the left boundaries.

9.5 Hilbert Curve [190152]

Hilbert index of (x, y) in $O(\log n)$, $n = 2^k$. Use it to order Mo's queries instead of block sorting.

```
ll hilbert(int n, int x, int y) {
    ll res = 0;
    for (int s = n / 2; s; s >>= 1) {
        int rx = (x & s) > 0, ry = (y & s) > 0;
        res += s * 1ll * s * ((3 * rx) ^ ry);
        if (!ry) {
            if (rx) x = s - 1 - x, y = s - 1 - y;
            swap(x, y);
        }
    }
    return res;
} // n = 2^k
```


9.6 DynamicConvexTrick* [673ffd]

Line container (CHT) with arbitrary insert order, amortised $O(\log n)$, keeps the **maximum. Integer coordinates only.**

// only works for integer coordinates!! maintain max

```
struct Line {
    mutable ll a, b, p;
    bool operator<(const Line &rhs) const {
        return a < rhs.a; }
    bool operator<(ll x) const { return p < x; }
};

struct DynamicHull : multiset<Line, less<>> {
    static const ll kInf = 1e18;
    ll Div(ll a, ll b) {
        return a / b - ((a ^ b) < 0 && a % b); }
    bool isect(iterator x, iterator y) {
        if (y == end()) { x->p = kInf; return 0; }
        if (x->a == y->a) x->p = x->b > y->b ?
            kInf : -kInf;
        else x->p = Div(y->b - x->b, x->a - y->a);
        return x->p >= y->p;
    }
    void addline(ll a, ll b) {
        auto z = insert({a, b, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x,
            y)) isect(x, y = erase(y));
        while ((y = x) != begin() &&
            (--x)->p >= y->p) isect(x, erase(y));
    }
    ll query(ll x) {
        auto l = *lower_bound(x);
        return l.a * x + l.b;
    }
};
```

Usage:

```
DynamicHull hull;
hull.addline(2, 3); hull.addline(-1, 10);
ll best = hull.query(x);
```

9.7 All LCS* [9542ce]

Returns every distinct longest common subsequence in lexicographic order. The result is directly testable and uses the standard $O(|s||t|)$ DP plus memoised reconstruction.

// Return every distinct longest common subsequence, in
// lexicographic order.

```
vector<string> all_lcs(const string &s,
    const string &t) {
    int n = (int)s.size(), m = (int)t.size();
    vector<vector<int>> dp(n + 1, vector<int>(m + 1));
    for (int i = n - 1; i >= 0; --i)
        for (int j = m - 1; j >= 0; --j)
            dp[i][j] = s[i] == t[j] ? 1 + dp[i + 1][j + 1] :
                max(dp[i + 1][j], dp[i][j + 1]);
    vector<vector<vector<string>>> memo(n + 1,
        vector<vector<string>>(m + 1));
    vector<vector<char>> seen(n + 1,
        vector<char>(m + 1));
    auto go = [&](auto &&self, int i,
        int j) -> vector<string> & {
        if (seen[i][j]) return memo[i][j];
        seen[i][j] = 1;
        auto &ret = memo[i][j];
        if (!dp[i][j]) return ret.push_back(""), ret;
        if (i < n && j < m && s[i] == t[j] &&
            dp[i][j] == 1 + dp[i + 1][j + 1]) {
            for (string x : self(self, i + 1,
                j + 1)) ret.push_back(s[i] + x);
        } else {
            if (i < n && dp[i + 1][j] == dp[i][j])
                for (string x : self(self,
                    i + 1, j)) ret.push_back(x);
            if (j < m && dp[i][j + 1] == dp[i][j])
                for (string x : self(self, i,
                    j + 1)) ret.push_back(x);
        }
        sort(ret.begin(), ret.end());
        ret.erase(unique(ret.begin(),
            ret.end()), ret.end());
        return ret;
    };
    return go(go, 0, 0);
}
```

9.8 AdaptiveSimpson* [95a73d]

Adaptive Simpson integration with Richardson extrapolation. Use eval2 to pre-split multi-peak integrands.

```
template<class Func, class d = double>
struct Simpson {
```

```
using pdd = pair<d, d>;
Func f;
pdd mix(pdd l, pdd r, optional<d> fm = {}) {
    d h = (r.first - l.first) / 2,
        v = fm.value_or(f(l.first + h));
    return {v, h / 3 * (l.second + 4 * v + r.second)};
} d eval(pdd l, pdd r, d fm, d eps) {
    pdd m((l.first + r.first) / 2, fm);
    d s = mix(l, r, fm).second;
    auto [flm, sl] = mix(l, m);
    auto [fmr, sr] = mix(m, r);
    d delta = sl + sr - s;
    if (abs(delta) <=
        15 * eps) return sl + sr + delta / 15;
    return eval(l, m, flm, eps / 2) +
        eval(m, r, fmr, eps / 2);
} d eval(d l, d r, d eps) {
    return eval({l, f(l)}, {r, f(r)},
        f((l + r) / 2), eps);
} d eval2(d l, d r, d eps, int k = 997) {
    d h = (r - l) / k, s = 0;
    for (int i = 0; i < k; ++i, l += h)
        s += eval(l, l + h, eps / k);
    return s;
}
};
```

template<class Func>

Simpson<Func> make_simpson(Func f) { return {f}; }

Usage:

```
auto f=make_simpson([](double x){return sin(x);});
double area=f.eval(0,acos(-1.0),1e-10); // area ≈ 2
```

9.9 Simulated Annealing [177cfb]

Generic minimising annealing routine: caller supplies a score and neighbour proposal, and receives the best state found.

// Generic minimising simulated annealing.

// 'next_state(state, rng)' proposes

// a neighbour and 'score(state)' is smaller for better

// states.

template<class State, class Score, class NextState>

```
State simulated_annealing(
    State current, Score score, NextState next_state,
    mt19937_64 &rng, int iterations = 100000,
    double temperature = 100000.0,
    double cooling = 0.99995) {
    State best = current;
    double cur = score(current), best_score = cur;
    uniform_real_distribution<double> unit(0.0, 1.0);
    for (int it = 0; it < iterations; ++it) {
        State candidate = next_state(current, rng);
        double cand = score(candidate);
        if (cand < cur || exp((cur - cand) /
            max(temperature, 1e-15)) >= unit(rng))
            current = candidate, cur = cand;
        if (cur < best_score) best = current,
            best_score = cur;
        temperature *= cooling;
    }
    return best;
}
```

Usage:

```
mt19937_64 rng(7122);
int best=simulated_annealing(0,[](int x){return abs(x-3);},
    [](int x,&auto&){return x<3?x+1:x-1;},rng); // best=3
```

9.10 Tree Hash* [f75ff7]

Order-independent rooted tree hash in $O(n)$ through rooted_tree_hash; for unrooted trees hash at each centroid and take the minimum.

using tree_hash_u64 = unsigned long long;

constexpr tree_hash_u64 tree_hash_default_seed =
 0x9e3779b97f4a7c15ULL;

```
tree_hash_u64 tree_hash_shift(tree_hash_u64 x) {
    x ^= x << 13, x ^= x >> 7, x ^= x << 17;
    return x;
}
```

```
tree_hash_u64 tree_hash_dfs(
    const vector<vector<int>> &g, int u, int parent,
    tree_hash_u64 seed) {
    if (!seed) seed = tree_hash_default_seed;
    tree_hash_u64 sum = seed;
    for (int v : g[u]) if (v != parent)
        sum += tree_hash_shift(tree_hash_dfs(g,
            v, u, seed));
    return sum;
}
```

tree_hash_u64 rooted_tree_hash(

```
const vector<vector<int>> &g, int root = 0,
tree_hash_u64 seed = tree_hash_default_seed) {
return tree_hash_dfs(g, root, -1, seed);
}
```

9.11 Binary Search On Fraction [7e5f4f]

Smallest fraction satisfying a caller-supplied monotone predicate, via Stern-Brocot descent with doubling; call `frac_bs(N, pred)`.

```
struct Q {
    ll p, q;
    Q go(Q b, ll d) { return {p + b.p*d, q + b.q*d}; }
};
// The predicate is supplied by the caller and must be
// monotone along the
// Stern--Brocot interval.
// returns smallest p/q in [lo, hi] such that
// pred(p/q) is true, and 0 <= p,q <= N
template<class Pred>
Q frac_bs(ll N, Pred pred) {
    Q lo{0, 1}, hi{1, 0};
    if (pred(lo)) return lo; assert(pred(hi));
    bool dir = 1, L = 1, H = 1;
    for (; L || H; dir = !dir) {
        ll len = 0, step = 1;
        for (int t = 0; t < 2 && (t ? step/=2 : step*=2);)
            if (Q mid = hi.go(lo, len + step);
                mid.p > N || mid.q > N || dir ^ pred(mid))
                t++;
            else len += step;
        swap(lo, hi = hi.go(lo, len));
        (dir ? L : H) = !!len;
    } return dir ? hi : lo;
}
Usage:
Q x=frac_bs(100,[](Q v){
return v.q==0 || 3*v.p>=2*v.q;
}); // x=2/3
```

9.12 Min Plus Convolution* [09b5c3]

$c_k = \min_{i+j=k} (a_i + b_j)$ by divide and conquer on monotone decisions. Requires a convex, barbitrary.

```
// a is convex a[i+1]-a[i] <= a[i+2]-a[i+1]
vector<int> min_plus_convolution(vector<int> &a,
vector<int> &b) {
    int n = SZ(a), m = SZ(b);
    vector<int> c(n + m - 1, INF);
    auto dc = [&](auto Y, int l, int r, int jl, int jr) {
        if (l > r) return;
        int mid = (l + r) / 2, from = -1, &best = c[mid];
        for (int j = jl; j <= jr; ++j)
            if (int i = mid - j; i >= 0 && i < n)
                if (best > a[i] + b[j])
                    best = a[i] + b[j], from = j;
        Y(Y, l, mid - 1, jl, from),
        Y(Y, mid + 1, r, from, jr);
    };
    return dc(dc, 0, n - 1 + m - 1, 0, m - 1), c;
}
Usage:
vector<int> a={0,1,4}, b={3,0}; // a must be convex
auto c=min_plus_convolution(a,b); // c={3,0,1,4}
```

9.13 Bitset LCS [41e379]

LCS in $O(nm/w)$ using subtract-and-borrow on 64-bit words; call `bitset_lcs(a,b)`.

```
int bitset_lcs(const vector<int> &a,
const vector<int> &b) {
    int m = (int)b.size(), W = (m + 63) >> 6;
    unordered_map<int, vector<unsigned long long>> mask;
    for (int j = 0; j < m; ++j)
        mask[b[j]].resize(W),
        mask[b[j]][j >> 6] |= 1ULL << (j & 63);
    vector<unsigned long long> f(W), x(W), y(W);
    for (int value : a) {
        auto it = mask.find(value);
        if (it == mask.end()) continue;
        unsigned long long carry = 1;
        for (int i = 0; i < W; ++i) {
            y[i] = (f[i] << 1) | carry;
            carry = f[i] >> 63, x[i] = f[i] | it->second[i];
        }
        unsigned long long borrow = 0;
        for (int i = 0; i < W; ++i) {
            unsigned long long rhs = y[i] + borrow;
            borrow = (rhs < y[i]) || (x[i] < rhs);
            f[i] = x[i] & ~(x[i] - rhs);
        }
        if (m & 63) f.back() &= (1ULL << (m & 63)) - 1;
    }
}
```

```
int ans = 0;
for (auto word : f) ans += popcount(word);
return ans;
}
```

9.14 N Queens Problem [df7da7]

Constructs one solution in $O(n)$ by cases on $n \bmod 6$ —no search. No solution exists for $n=2,3$.

```
void solve(vector<int> &ret, int n) {
// no sol when n=2,3
if (n == 2 || n == 3) return;
if (n % 6 == 2) {
    for (int i = 2; i <= n; i += 2) ret.pb(i);
    ret.pb(3); ret.pb(1);
    for (int i = 7; i <= n; i += 2) ret.pb(i);
    ret.pb(5);
} else if (n % 6 == 3) {
    for (int i = 4; i <= n; i += 2) ret.pb(i);
    ret.pb(2);
    for (int i = 5; i <= n; i += 2) ret.pb(i);
    ret.pb(1); ret.pb(3);
} else {
    for (int i = 2; i <= n; i += 2) ret.pb(i);
    for (int i = 1; i <= n; i += 2) ret.pb(i);
}
}
Usage:
vector<int> column;
solve(column, 8); // column[row] is 1-based; empty for n=2,3
```

9.15 Digit DP [baeb8a]

Binary digit DP compressed to Fibonacci counts; `calc(x)` counts values in $[0, x]$ with no adjacent one bits.

```
// Number of integers in [0, x] whose binary
// representation has no adjacent
// one bits. This is a compact, reusable digit-DP
// example.
int calc(int x) {
    if (x < 0) return 0;
    int ways[31] = {1, 2}, ans = 0, prev = 0;
    for (int i = 2; i <= 30; ++i) ways[i] = ways[i - 1] + ways[i - 2];
    for (int i = 30; i >= 0; --i) {
        if (x >> i & 1) {
            ans += ways[i];
            if (prev) return ans;
            prev = 1;
        } else prev = 0;
    }
    return ans + 1;
}
Usage:
int answer=calc(R)-calc(L-1); // range [L,R]
```

10 Python

10.1 Misc

Importable Python fallbacks for exact decimals/ fractions, matrix and fast-array parsing, modular powers, seeded random helpers, and formatting. Huge integer input raises Python 3.11+’s digit limit.

```
from decimal
import Context, Decimal, MAX_EMAX, MAX_PREC, ROUND_FLOOR, setcontext
from fractions import Fraction
from random import Random
import sys
if hasattr(sys
, "set_int_max_str_digits"): sys.set_int_max_str_digits(5_000_000)
def decimal_multiply(a: str, b: str) -> Decimal:
    setcontext
    (Context(prec=MAX_PREC, Emax=MAX_EMAX, rounding=ROUND_FLOOR))
    return Decimal(a) * Decimal(b)
def limit_fraction(value: str, max_denominator: int) ->
    Fraction: return Fraction(value).limit_denominator(max_denominator)
def read_matrix(lines: list[str], n: int, m: int) -> list[list[int]]:
    matrix = [list(map(int, line.split())) for line in lines[:n]]
    if len(matrix) != n or any(len(row) != m for row in matrix):
        raise ValueError("matrix dimensions do not match n x m")
    return matrix
def modular_power
(a: int, b: int, modulus: int) -> int: return pow(a, b, modulus)
def random_examples
(rng: Random, values: list[int]) -> tuple[int, int, list[int]]:
    if not values:
        raise ValueError("values must be non-empty")
    shuffled = values.copy()
    rng.shuffle(shuffled)
    return rng.randint(1, 10), rng.choice(values), shuffled
def format_example(num: int, pi: float
, text: str) -> str: return f"num: {num}, pi: {pi:.2f}, str: {text}"
def parse_int_array(data: bytes) -> tuple[int, list[int]]:
    tokens = data.split()
    if not tokens:
        raise ValueError("missing array length")
    n = int(tokens[0])
    if len(tokens) != n + 1:
        raise ValueError("array length does not match input")
    return n, list(map(int, tokens[1:]))
```